Fairgame

The Decentralized Autonomous Propfirm

A formal specification of a self-balancing, verifiable proprietary-trading protocol on Sonic

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Abstract. We specify Fairgame, a decentralized autonomous proprietary-trading firm deployed on Sonic, an EVM-compatible layer-1 blockchain chosen for its fast finality and low fees. Traders buy evaluations in stablecoins, trade a non-transferable unit of account, Sonic Virtual USD (svUSD), against oracle prices in a fully on-chain simulator, and, once they pass, operate funded svUSD accounts whose profits are paid out in USDC from a transparent reserve. Conventional prop firms stay solvent through opaque pricing, discretionary payout denial and identity checks, and their customers can verify neither payout decisions nor reserves; firms marketed as crypto or decentralized prop firms often change only the payout rail. Fairgame makes its rules, account states, reserve balances, obligations and payouts, and every price it uses, verifiable on-chain at every block; since none of the conventional levers is available to a permissionless protocol, it derives its prices from first principles instead. We prove that, under a fair-execution discipline and explicit assumptions on oracle prices and keepers, every trader without an informational edge faces a supermartingale, up to one stated gap for conditional orders. Optional stopping then yields closed-form upper bounds on the probability of passing an evaluation and on the lifetime payout of any funded account. These bounds define a *farm-proof price* at or above which no strategy without an informational edge, including arbitrary coalitions of Sybil accounts, has positive expected value, and a liability measure that can be evaluated at every block. On this valuation layer we build a feedback controller that adjusts prices, the menu of offered rule sets and funded capacity from realized cash flows and live coverage; we establish the stability region of its price loop and identify the adverse-selection condition under which price stops being an effective lever. We further specify a priority fee waterfall, a stress-coverage invariant that bounds exposure to correlated winners, solvent-by-construction tournaments for bootstrapping, optional proof-of-personhood discounts with bounded exposure, a verifiable trader passport, and a copy-capital vault that turns statistically skilled traders from a liability into a source of revenue.

All parameters in this paper are proposals subject to calibration. Nothing in this document is an offer of any product, investment advice or legal advice.

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1 Introduction

1.1 The evaluation business model

A retail *proprietary trading firm* ("prop firm") sells *evaluations*. A trader pays a fee, receives a simulated account of nominal size S, and must reach a profit target before breaching a set of loss limits. Traders who pass receive a *funded* account, in most firms also simulated, and are paid a share α of the profits it records, typically $\alpha \in [0.7, 0.9]$; advertised splits reach 90% [3].

Economically, the firm writes path-dependent contingent claims on the performance of its customers. The fee is a premium; the payout is a claim with capped downside for the trader and uncapped upside. The firm is solvent when premia exceed the expected value of claims plus operating costs. Conventional firms maintain this inequality with three levers that are largely invisible to their customers:

(i) **Behavioral under-performance.** Evaluations are typically priced below the expected payout that a *disciplined random* strategy would earn (Section 5 makes this quantity precise). The model works because most customers perform worse than random once costs, overtrading and time pressure are accounted for.

(ii) **Discretion.** Payouts can be denied after the fact under loosely specified rules on "gambling", "consistency" or "prohibited strategies", whose scope the firm determines at its own discretion [24, 25, 60].

(iii) **Identity.** Know-your-customer checks and the linking of accounts through shared devices and payment details make it hard to run many accounts, and with them the systematic harvesting of under-priced evaluations.

None of these levers survives in a permissionless setting. Bots execute disciplined random strategies at scale; code that exercises discretion is a trusted party in disguise; and wallets are free. A decentralized prop firm that copies conventional pricing is therefore exploitable by construction.

The conventional model is also opaque to its customers. Published terms can make every payout subject to the firm's final approval after a compliance review [25]. Belgium's financial regulator has warned that, because the firm alone decides the conditions of use, consumers find it difficult to determine whether they are owed anything and, if so, how much [23], and Italy's securities regulator has received complaints that promised profit shares were not paid [15]. Customers generally cannot see the reserves from which they are paid, or tell whether the firm will keep operating: in February 2024, several firms suspended trading after losing access to their trading platform, one deferring all payouts until service could be restored [44, 45]. Firms marketed as crypto or decentralized prop firms often change only the payout rail: a payout platform settles stablecoin transfers from funds the firm provides [40], while evaluation, account state and rule enforcement stay off-chain and reserves go unpublished. Some entrants that pay out on-chain still run evaluation on their own systems [33]. In Fairgame, funds are held in on-chain vaults that anyone can audit (Section 6); payouts are computed by a published formula from on-chain account state and paid from a transparent reserve (Algorithm 3); claims are released within a bounded window and can be challenged only with objective evidence (Section 4.6); and solvency and coverage conditions are enforced on-chain whenever risk is added (Sections 7.7 and 7.8). Solvency thus rests on bounds and invariants that anyone can check rather than on an operator's discretion, but only under the assumptions stated in Sections 5 and 7.8: Section 12.3 lists the design questions that remain open, and Section 12 analyzes contract and oracle risk.

1.2 This paper

We specify *Fairgame*,[1] the Decentralized Autonomous Propfirm, a protocol that runs on Sonic, an EVM-compatible layer-1 blockchain with fast, deterministic finality [50]. Traders purchase evaluations in stablecoins, trade a non-transferable unit of account, *Sonic Virtual USD* (svUSD), against oracle prices in a fully on-chain simulator, and are paid in USDC from a transparent reserve when their funded accounts record profits. Every price, rule and capacity limit in Fairgame is either derived from a valuation theory that holds against every strategy without an informational edge, or set by an on-chain feedback controller within hard bounds. Nothing is discretionary.

Design goals.

G1. Verifiability. Every state transition is reproducible from on-chain data and signed oracle reports.

G2. Solvency without discretion. Payouts are a deterministic function of state; solvency follows from pricing and invariants, not from a right to refuse payment.

G3. Permissionless access. Identity is optional; the protocol is safe against arbitrary numbers of anonymous accounts.

G4. Self-balancing economics. Prices, the menu of rules and funded capacity respond automatically to cash inflows, cash outflows and live liabilities.

G5. Trader fairness. Rules are frozen at purchase, published in full and enforced identically for everyone.

G6. Sustainability. Revenue is routed through a priority waterfall into a payout reserve, an insurance fund, a sustainability fund and development, owned and governed by the community.

G7. Sonic-native. The design relies on Sonic's fast finality for short execution latency and keeper windows, and hence narrow spreads, on its low fees and fee monetization for frequent engine transactions and gas sponsorship, and on native USDC for payouts (Section 2.3).

Contributions.

(1) A *fair-execution discipline* for oracle-priced simulated trading: deterministic first-report fills after a commit delay, freshness-bounded conditional orders and a spread condition. Under it and explicit assumptions on oracle prices and keepers, every trader without an informational edge faces a supermartingale (Theorem 5.4), up to one gap for conditional orders that we state as an open question.

(2) A *zero-skill valuation theory*: closed-form bounds on the probability of passing an evaluation (Theorem 5.6) and on the lifetime payout of a funded account (Theorem 5.7). They yield the zero-skill value $V^\star(R)$ and the *farm-proof price* $F^\star(R)$ of a rule set R, show that no coalition of zero-skill Sybil accounts profits in expectation at or above that price (Theorem 5.13), and give a liability measure that can be evaluated at every block (Proposition 5.15).

(3) A *self-balancing controller* that adjusts prices, the offered rule menu and funded capacity from coverage and loss-ratio signals, with an explicit stability region for its price loop (Propositions 7.2 and 7.3) and the adverse-selection condition under which price ceases to be an effective lever (Proposition 7.4).

[1] The name refers to the classical notion of a fair game, that is, a martingale. Theorem 5.4 shows that, under fair execution, trading on Fairgame is at best a fair game for a trader without an informational edge: their account equity, counting profits already claimed, is a supermartingale, with the qualification for conditional orders stated after the theorem.

(4) A *treasury architecture*: a priority fee waterfall, a stress-coverage invariant that bounds the reserve's exposure to correlated winners whenever risk is added (Theorem 7.6), and a set of machine-checkable protocol invariants (Section 12.2), with the engine-design questions that these results leave open or take as assumptions (Section 12.3).

(5) *Extensions* that preserve these guarantees: solvent-by-construction tournaments for bootstrapping (Proposition 9.2), optional proof-of-personhood discounts with bounded exposure (Proposition 8.1), a verifiable trader passport, and a copy-capital vault that converts statistically skilled traders from a liability into revenue (Proposition 10.1).

1.3 Related work

Oracle-priced derivatives. Perpetual-futures protocols such as GMX [26] settle trades against oracle prices and use keeper-executed, two-step orders to neutralize trading against stale prices and front-running [17]; the Perps V2 markets of Synthetix used the same pattern, with a minimum delay between an order and its execution [59]. Fairgame adopts the two-step pattern but differs structurally: there is no counterparty pool and no real position. Trades are simulated, and the protocol's only real liability is the payout function. Execution rules can therefore be chosen purely for their effect on the trader's information set, which Section 5 exploits.

On-chain prop firms. Several firms launched in 2025–2026 publish payout contracts, reserve addresses or rule hashes on public chains; Propr, for example, pays out in USDC on-chain and publishes every payout transaction [33]. In the designs we reviewed, the trading engine typically runs off-chain or under operator control, and evaluations are sold from fixed price lists. Fairgame puts the engine itself on-chain, derives prices from first principles and replaces fixed pricing with feedback control.

Insurance economics and control. The controller borrows from actuarial practice (loss ratios, capacity limits, reserving) and from feedback control [2]. Its failure mode under adverse selection is the classical one [1, 41].

Tournaments and performance evaluation. Rank-order incentives [28] can lead trailing competitors to take on more risk [7], which shapes our tournament scoring. Skill detection under multiple testing uses the probabilistic and deflated Sharpe ratios [4, 5].

Foundations. The valuation theory rests on optional stopping for supermartingales [62] and its classical gambler's-ruin corollary [22], applied under the martingale null of efficient prices [21, 42]. Without a logically centralized authority that certifies identities, Sybil attacks are always possible except under extreme and unrealistic assumptions [19], so Fairgame treats Sybil resistance economically rather than through identity. The presentation of the state machine follows the yellow-paper convention of [63].

1.4 Organization

Section 2 fixes the model, Section 3 gives an overview of the protocol and Section 4 specifies the trading engine. Section 5 develops the valuation theory. Sections 6 and 7 specify the treasury and the controller. Sections 8 to 10 cover identity and reputation, tournaments and the copy-capital vault. Sections 11 to 13 treat governance, security and the implementation on Sonic, and Sections 14 and 15 discuss regulatory considerations and the roadmap. The appendices collect notation, worked examples and longer proofs.

2 Model and preliminaries

2.1 Conventions

We write $(x)^+ = \max(x, 0)$, $x \wedge y = \min(x, y)$ and $\mathbf{1}_A$ for the indicator of an event A. Monetary amounts are real numbers. Amounts in svUSD and USDC share a numerical scale so that rules read naturally (an account with $S = 25{,}000$ is "a \$25k account"), but the two are never exchanged: the only map from svUSD to USDC is the payout function of Section 4.6. A notation table is given in Appendix A.

2.2 Execution environment

Sonic is an EVM-compatible proof-of-stake layer-1 blockchain with fast, deterministic finality, whose native token S pays for gas [50, 54]. Four platform features matter for Fairgame.

- **Fee Monetization** (FeeM) currently pays a registered application, in S, 90% of the network fees for the gas consumed in its contracts; the other 10% goes to validators [52]. Sonic Labs said in February 2026 that it is evaluating a tiered structure [47], so Fairgame treats FeeM as upside rather than as a load-bearing revenue line.
- **Native stablecoin rails.** Circle has issued USDC natively on Sonic, with its cross-chain transfer protocol (CCTP V2), since May 2025 [13], giving payouts a regulated settlement asset that moves between chains without a third-party bridge. USSD, a Treasury-backed stablecoin built on Frax's frxUSD infrastructure that Sonic Labs describes as native to the Sonic ecosystem [58], is a candidate secondary fee asset.
- **Low-latency pull oracles.** Pyth [34, 35] and Chainlink Data Streams [12] are both deployed on Sonic, so prices from two independent oracle networks can be verified on-chain at the moment they are used.
- **Protocol-level gas sponsorship.** Since release v2.1.2 an application can pay its users' gas from an on-chain deposit matched to the contracts or functions it chooses, and release v2.2, live since September 22, 2026, can charge that deposit only for transactions that succeed [49, 56, 57].

We model time as continuous, $t \geq 0$. Transactions are totally ordered by the chain, and each carries the timestamp of its block, which the model takes as the time of the transaction. Sonic's block timestamps are whole seconds; we assume that each trails the real inclusion time by less than one second. An order recorded at $\hat{t}_c$ is therefore decided before $\hat{t}_c + 1\,\text{s}$, while the engine fills it no earlier than $\hat{t}_c + \delta$, so the results below apply on chain with $\hat{t}_c + 1\,\text{s}$ as the commit time and $\delta - 1\,\text{s}$ as the commit delay, for market orders and for cancellations alike. In particular, condition (A3) of Section 5 becomes $\delta \geq \Delta_o + 1\,\text{s}$, the calibration rule of Section 13.2.

2.3 Why Sonic

Fairgame runs its whole engine on-chain: every commit, settlement, conditional execution and breach proof is a transaction. The chain therefore fixes how quickly an action takes effect and what it costs. Speed is the main reason Fairgame is built on Sonic.

Speed. Sonic reaches consensus with a proof-of-stake, asynchronous Byzantine-fault-tolerant (aBFT) protocol on a directed acyclic graph of validator events, a continuation of the Lachesis protocol of Fantom Opera [30, 48], so finality is deterministic rather than probabilistic. Sonic Labs states that transactions are irreversible in under a second [50]; its consensus documentation puts the time from submission to inclusion in the main chain at about 1–2 s [48]. For the market-wide liquidations that began on October 10, 2025, Sonic Labs reports an average of 0.58 s per block and a peak of 1,465 transactions per second, against a stated capacity of 400,000 [50,

55]. As an independent reference, Circle issues attestations for standard CCTP transfers only after what it calls hard finality: it requires one block confirmation for transfers from Sonic, with attestations available after about 8 s on average, against about 65 Ethereum blocks (15–19 minutes) for transfers from Ethereum and from the Arbitrum, Base and OP Mainnet rollups [14]. These are one bridge operator's attestation requirements rather than measurements of finality, but they show that Circle treats a Sonic transaction as final after a single block. The parameters of Table 6 assume only that a transaction is final within one to two seconds.

What speed buys. Fairness does not come from speed: on any chain, a market order fills at the first report published at or after $t_c + \delta$, which must be admissible (Definition 4.4), and Theorem 5.4 holds wherever δ, Δ_f and the spreads are chosen to satisfy its assumptions. Speed determines what this discipline costs.

(i) *Latency.* A fill follows the trader's decision by the inclusion time, δ and the wait for the next report; Pyth states that its feeds update every 400 ms [38]. With $\delta = 2$ s the whole delay is a few seconds, and since an account holds at most one pending market order, this delay plus the settlement also bounds how soon the next order can be committed.

(ii) *Keeper windows and spreads.* A conditional order executes only on a report at most Δ_f old (Section 4.3), so Δ_f must exceed the time a keeper needs to land a transaction, and the base spread must cover the selection value $v_j(\Delta_f) \approx \sigma_j\sqrt{2\Delta_f/\pi}$ (Proposition 5.3). Fast inclusion makes $\Delta_f = 2$ s feasible; a window ten times longer would need a spread floor $\sqrt{10} \approx 3.2$ times wider.

(iii) *Breach proofs.* A proof must use reports published after the account's last settlement (Algorithm 2), so a breach from which the account recovers before it next settles can no longer be proven. Landing proofs within a second or two of the breaching report keeps the effective monitoring interval Δ_m, and with it the overshoot ξ priced into the effective loss limit (Proposition 5.14), at the few seconds assumed in calibration.

(iv) *Current state.* Keepers are paid per settlement, so fills, and with them the positions and liabilities behind coverage (Proposition 5.15 and Invariant I1), are recorded seconds after their reports.

Fees. Each market order costs a commit and a settlement that verifies signed reports from two oracle networks, on top of triggers, breach proofs and epoch updates. Sonic's base fee has a floor of 50 gwei (5×10^{-8} S per unit of gas) and rises only when more than half of the network's gas budget is in use [53]. Sonic Labs reports that it held at that floor through the October 2025 liquidations, although voluntary tips raised the average price paid about sixfold on October 10 [55]. At the floor, a transaction that uses one million gas costs 0.05 S, so keeper rewards and gas sponsorship are cheap and predictable, also in volatile markets, when triggers and breach proofs are most frequent.

Other reasons. Sonic supports the EVM feature set of Ethereum's Cancun upgrade except blob transactions and, since release v2.1, the Prague set including EIP-7702 [51]; ERC-4337 EntryPoint contracts are deployed with commercial bundler and paymaster services [32]. With the platform features listed above, Fairgame is built from standard Ethereum contracts and tooling.

Trade-offs. Like any asynchronous BFT protocol, Sonic guarantees safety only while validators holding less than one third of the stake are faulty [30]; a validator must lock at least 500,000 S [54]. A halt of the chain delays every transition but changes few outcomes: market orders and breach proofs use reports selected by publish time and resolve once the chain resumes, whereas conditional orders need fresh reports, so for them a halt acts like a price gap (Table 4). FeeM may change, so

the results of Sections 5 and 7 do not rely on it (Section 6). Several blocks can share one second, but block timestamps are whole seconds, which the timing rules must respect (Section 13). Finally, Sonic's mainnet has run only since December 2024 [46].

2.4 Markets and oracle observations

Let $\mathcal{M} = \{1, \dots, M\}$ be the set of listed markets, each a liquid crypto-asset quoted in USD.

Definition 2.1 (Observation)**.** An *observation* of market j is a signed oracle report $\omega = (j, P, c, u, u^-)$ with price $P > 0$, confidence half-width $c \geq 0$, publish time u, and the publish time $u^- < u$ of the immediately preceding report of the same feed.

Definition 2.2 (Admissibility)**.** An observation ω processed at time t is *admissible*, written $\mathsf{Adm}_t(\omega)$, if (i) its signature verifies against the registered primary oracle; (ii) its relative confidence is bounded, $c/P \leq \kappa_{\max}$; (iii) a secondary-oracle observation ω' of the same market with $|u' - u| \leq \Delta_{\text{sync}}$ is supplied and agrees, $|P - P'|/P \leq \varepsilon_x$; and (iv) where freshness is required, $t - u \leq \Delta_f$.

Condition (iii) limits what a compromise of a single oracle network can do: it can move admissible prices, and with them simulated equity, by at most about ε_x (Section 12). Condition (iv) is imposed on conditional orders (Section 4.3); market orders and breach proofs use the stronger timing rules stated there.

Assumption 2.3 (Bounded oracle latency)**.** There is $\Delta_o \geq 0$ such that each oracle price $P_j(u)$ is determined by public market information available up to time $u - \Delta_o$.

2.5 Information, prices and skill

Fix a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$, where $\mathbb{F} = (\mathcal{F}_t)_{t \geq 0}$ is the *public information filtration*: everything observable by any market participant at time t, including prices on external venues. Define the *oracle filtration* $\mathbb{G} = (\mathcal{G}_t)_{t \geq 0}$ by $\mathcal{G}_t = \mathcal{F}_{t - \Delta_o}$, with $\mathcal{F}_s = \mathcal{F}_0$ for $s < 0$. By Assumption 2.3, every oracle price process P_j is $\mathbb{G}$-adapted, while a trader's information at time t is $\mathcal{F}_t = \mathcal{G}_{t + \Delta_o}$: a fast trader can be up to Δ_o ahead of the oracle. We assume in addition that oracle publish times are $\mathbb{G}$-stopping times, as they are under a fixed publication schedule.

Assumption 2.4 (Martingale null)**.** On the horizons relevant to an account, each oracle price P_j is a $\mathbb{G}$-martingale.

Assumption 2.4 is the efficient-markets null [21, 42]: given what the oracle has incorporated, the best forecast of future oracle prices is the current one. It is not a claim that markets are efficient; it is the benchmark against which Fairgame prices its liabilities.

Definition 2.5 (Zero-skill and skilled strategies)**.** A trading strategy is *zero-skill* if every order it commits at time t is $\mathcal{F}_t$-measurable, that is, uses only public information. A trader is *skilled* if her information filtration $\mathbb{H} \supsetneq \mathbb{F}$ admits forecasts of oracle prices with non-zero conditional drift.

Zero-skill strategies include every technical, momentum and martingale-betting system based on public data, run on any number of wallets in any combination. Skill is an informational or predictive edge that the market has not yet priced. Fairgame exists to find and pay it.

Remark 2.6 (Drift). Over days, crypto drift is small relative to volatility but not zero, and in trending regimes many zero-skill strategies with the same directional exposure win together. Fairgame does not price drift into evaluations. It treats aggregate directional exposure as a *correlated liability* and bounds it with stress capital and open-interest limits (Section 7.8).

2.6 Units of account

Definition 2.7 (svUSD)**.** *Sonic Virtual USD* (svUSD) is the unit of account in which simulated balances, positions and profits are recorded. An amount of svUSD exists only as a field of an account record, with the following properties:

(S1) **Non-transferable.** No interface moves svUSD between accounts or addresses; there is no token contract.

(S2) **Non-redeemable.** svUSD is never exchanged for any asset. Payouts in USDC are computed from recorded profits by the payout function and are *performance rewards*, not redemptions.

(S3) **Engine-only mutation.** Balances change only through engine transitions: fills, fees, claims and phase resets.

Property (S1) removes a whole class of attacks, such as moving balances between accounts or using svUSD as collateral elsewhere. Property (S2) avoids the economic and legal character of a redeemable token (Section 14).

3 Protocol overview

3.1 Actors

- **Traders** buy evaluations, trade svUSD accounts and claim payouts. They may act through smart accounts with scoped session keys (Section 13.4).
- **Keepers and watchers** settle orders, execute conditional orders, prove rule breaches and trigger epoch updates. All of these roles are permissionless and paid per action.
- **Oracles** publish signed price reports that the engine verifies on-chain.
- **Governance** maintains the catalog of rule sets, market listings and controller parameters within immutable bounds (Section 11).
- **Liquidity providers** (in a later phase) deposit USDC into the copy-capital vault (Section 10).

3.2 Account lifecycle

Figure 1 shows the lifecycle of an account. An account is minted in phase EVAL_1 when a trader buys an evaluation, and its rule set is frozen at that moment. It advances through the evaluation phases by meeting their profit targets, enters FUNDED (or the funding queue, if funded capacity is exhausted), and records claims until it breaches a loss limit or is retired.

3.3 Value flows

Figure 2 shows how value and information move through the protocol. Evaluation fees enter through the eval shop at the price set by the risk controller and are split by the fee router. Traders commit orders to the trading engine, which records fills against verified oracle reports. Funded accounts convert recorded profits into claims, and the payout manager pays the profit share in USDC from the payout reserve. The risk controller reads live equity, open interest and reserve balances and sets the price multiplier, the menu of rule sets on sale and funded capacity.

3.4 Design principles

P1. Rules are immutable per account. Parameters may change for new accounts only.

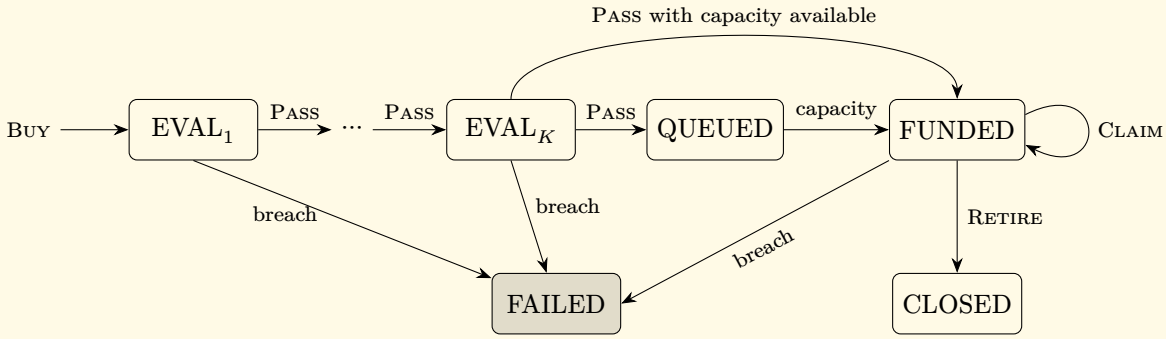


Figure 1: Account lifecycle. Rules are frozen at BUY. Breaches are proven permissionlessly (Algorithm 2). A passing account enters the funding queue only when funded capacity is exhausted (Section 7.7).

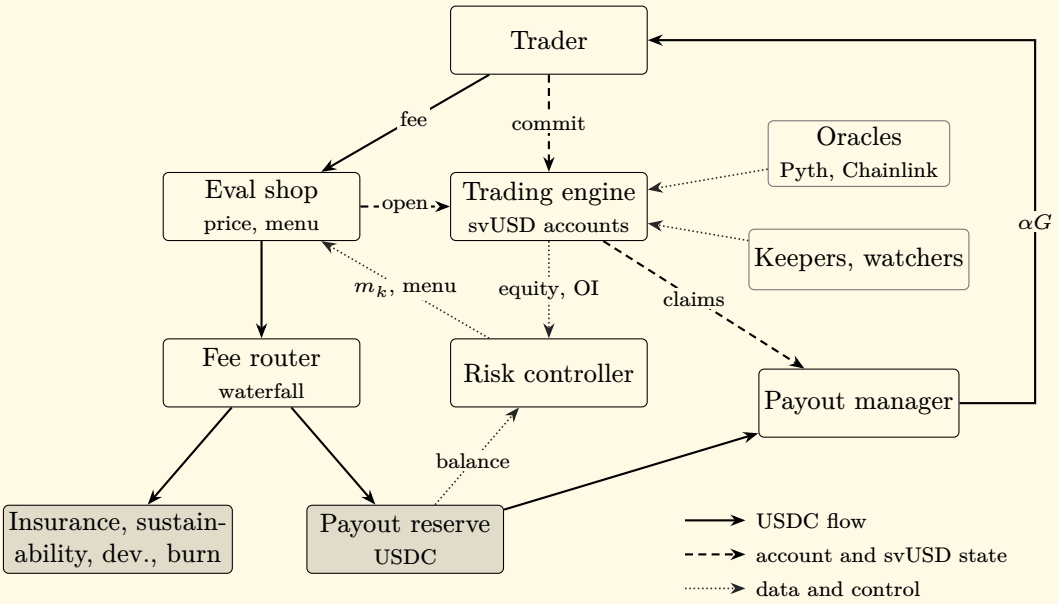


Figure 2: Value and information flows. Extensions (tournaments, identity, copy-capital vault) are omitted for clarity.

P2. Trades are against the oracle. There is no counterparty, no order book and no internal matching, and hence no wash trading, self-matching or liquidity games.

P3. Every liability is priced by its zero-skill value (Section 5).

P4. No discretion. Payouts are a deterministic function of on-chain state.

P5. Feedback over prediction. Aggregate signals drive parameters, within hard bounds (Section 7).

4 The trading engine

4.1 Rule sets

Definition 4.1 (Rule set)**.** A *rule set* is a tuple

$$R = (S,\ K,\ (T_k, D_k)_{k=1}^{K},\ L,\ S_f,\ D_f,\ L_f,\ \alpha,\ \lambda,\ n_{\text{pos}},\ g_{\min},\ \rho,\ \mathcal{M}_R)$$

where S is the evaluation balance; $K \in \{1, 2\}$ the number of evaluation phases; T_k and D_k the profit target and static maximum loss of phase k, as fractions of S; L the daily loss limit as a

fraction of S; S_f, D_f and L_f the funded balance, maximum loss and daily loss limit; $\alpha \in (0,1)$ the profit split; λ the leverage cap; n_{pos} the maximum number of simultaneously open markets; $g_{\min}$ the minimum claim as a fraction of S_f; $\rho \in [0,1]$ the fraction of the fee refunded with the first claim; and $\mathcal{M}_R \subseteq \mathcal{M}$ the tradable markets. The governance-approved *catalog* of rule sets is $\mathcal{R}$.

An account's rule set is fixed when it is bought (principle P1, Invariant I4). Minimum trading days, consistency rules and bans on particular strategies are deliberately absent: Proposition 5.12 shows that they are unnecessary for solvency, and they are the rules traders most often experience as arbitrary. Minimum holding times are likewise unnecessary, because execution is fair by construction (Section 5.1); an implementation may still impose one as defense in depth.

4.2 Accounts and equity

An account $a \in \mathcal{A}$ is an ERC-721 token locked to its owner (ERC-5192 [18]) that stores

$$a = (R_a,\ \text{ph}_a,\ S_a,\ B_a,\ (q_{a,j}, \bar{p}_{a,j})_{j\in\mathcal{M}},\ B^0_a,\ d_a,\ W_a,\ t_a,\ \mathsf{esc}_a).$$

Here $\text{ph}_a \in \{\text{EVAL}_1, \dots, \text{EVAL}_K, \text{QUEUED}, \text{FUNDED}, \text{FAILED}, \text{CLOSED}\}$ is the phase; $S_a \in \{S, S_f\}$ the reference balance of the current phase; B_a the realized balance; $q_{a,j}$ the signed position in units of asset j and $\bar{p}_{a,j}$ its average entry price; B^0_a the balance at the start of UTC day d_a; W_a the cumulative gross claims; t_a the time of the last settlement; and esc_a the account's USDC escrow for keeper rewards and bounties.

Definition 4.2 (Equity, notional, leverage and floors)**.** Given prices $P = (P_j)_{j\in\mathcal{M}}$, the *equity*, *notional* and *leverage* of a are

$$E_a(P) = B_a + \sum_j q_{a,j}\,(P_j - \bar{p}_{a,j}) - A_a, \qquad N_a(P) = \sum_j |q_{a,j}|\,P_j, \qquad \text{Lev}_a = N_a/E_a,$$

where $A_a \geq 0$ is accrued, unsettled crowding fees (Section 4.4). In a phase with maximum loss $D \in \{D_k, D_f\}$ and daily limit $L^\circ \in \{L, L_f\}$, the *static floor* and the *daily floor* are

$$f_a = S_a(1-D), \qquad f^{\text{day}}_a = B^0_a - L^\circ S_a.$$

The day-start balance B^0_a can be computed lazily: realized balances change only at settlements, so B^0_a equals the balance after the last settlement before the start of the day.

4.3 Orders and fair execution

Execution is where a simulator leaks value: any fill better than the market grants a free edge, and Section 5 shows that the entire pricing theory depends on there being none. Fairgame therefore separates the *decision* to trade from the *price* at which the trade is recorded.

Definition 4.3 (Order)**.** An order is $\mathsf{o} = (a, j, \Delta q, \kappa, p^{\text{trg}}, t_c)$ with account a, market $j \in \mathcal{M}_{R_a}$, signed size Δq, kind $\kappa \in \{\mathsf{MKT}, \mathsf{LMT}, \mathsf{STP}\}$, trigger price p^{trg} (conditional orders only) and commit time t_c, the timestamp of the COMMIT transaction.

Definition 4.4 (Deterministic fill time)**.** The fill observation of a market order committed at t_c is the *first* observation of market j published at or after $t_c + \delta$:

$$u^\star(\mathsf{o}) = \min\left\{u :\ \omega_j(u) \text{ published},\ u \geq t_c + \delta\right\}.$$

It is certified on-chain by the condition $u^- < t_c + \delta \leq u$ on the submitted report, which must also be admissible (Algorithm 1).

The fill of a market order is thus fixed at commit time, up to the oracle's future output. Market orders cannot be canceled. The SETTLE transaction that records the fill can be sent by anyone at any later time (*lazy settlement*). It cannot change the fill time or the fill price, and the risk checks are arranged so that it cannot change the admission decision either (Section 4.5), except through the choice of synchronized reports for the account's other markets, which matters only for an order at the margin of the leverage cap. Apart from this residual, no party, including a trader who also runs a keeper, holds an option after committing. While a market order is pending, the account cannot commit another one, and any transition that applies a report published after the pending order's fill time settles that order first, so reports are applied to each account in time order.

The engine specifies no outcome when the certified first report is not admissible, for instance because its confidence exceeds $\kappa_{\max}$ or no agreeing secondary report was published within Δ_{sync} of it. The certificate accepts only that one primary report, so SETTLE can then never succeed: the order stays pending, and because every transition that applies a later report must settle it first, the account can no longer trade, pass or claim, and a breach can be proven only from reports published before the order's fill time. The bounds of Section 5 are unaffected, since a blocked account can neither pass nor claim, but the account is frozen, and an adversary that controls the secondary network can freeze accounts by withholding reports (Section 12). The rule for this case is an open design question (Section 12.3).

Conditional orders. Limit and stop orders, including take-profit and stop-loss orders as reduce-only variants, are executed by TRIGGER with an observation ω that is admissible, fresh ($t - u \le \Delta_f$), published at or after $t_c + \delta$, and satisfies the trigger: $P \le p^{\text{trg}}$ for a buy limit or a sell stop, and $P \ge p^{\text{trg}}$ for a sell limit or a buy stop. Limit orders fill at p^{trg} and stop orders at P; both pay the half-spread of Section 4.4. A cancellation of a conditional order takes effect δ after it is committed, so it cannot exploit information the oracle has not yet incorporated.

Assumption 4.5 (Keeper liveness)**.** Whenever a conditional order is executable and the oracle networks' report-distribution services are available, some keeper executes it within Δ_f.

Keepers are permissionless and are paid per execution from the account's escrow, so Assumption 4.5 is an economic assumption about a competitive keeper market, backstopped by a protocol-operated keeper. It limits the residual choice of observation to a window of length Δ_f, whose value is bounded in Proposition 5.3. Every keeper, the backstop included, needs credentials (API keys) from both oracle providers to obtain reports (Section 12), so the backstop covers a shortage of keepers but not a provider-wide outage or loss of access. During such an outage no conditional order executes and no breach can be proven; afterwards, conditional orders execute on fresh reports that satisfy their triggers, so stop fills can slip and losses can overshoot floors by more than Proposition 5.14 prices. Theorem 5.4, which uses Assumption 4.5 through condition (A6), therefore assumes that both services are available while an account holds conditional orders, and the slippage and overshoot an outage causes are a residual left to the insurance fund (Table 4).

4.4 Execution prices and costs

For a fill of signed size Δq against observation $\omega = (j, \hat{P}, c, u, u^-)$, the half-spread is

$$s_j(\Delta q, \omega) = s_{0,j} + \kappa_c \frac{c}{\hat{P}} + \frac{|\Delta q|\ \hat{P}}{\Lambda_j}, \tag{1}$$

the sum of a base spread, a term that widens with oracle uncertainty, and a linear price-impact term with virtual depth Λ_j (in USD). A market or stop fill is recorded at $P^{\text{fill}} = \hat{P}\,(1 + \text{sgn}(\Delta q)\, s_j)$ and a limit fill at $p^{\text{trg}}(1 + \text{sgn}(\Delta q)\, s_j)$. A trading fee $\varphi_j\, |\Delta q|\, P^{\text{fill}}$ is debited from B_a.

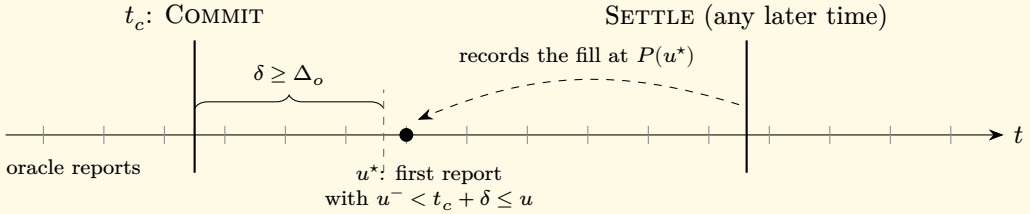


Figure 3: Two-step deterministic execution of a market order. The decision is committed at t_c; the price is that of the first oracle report published at or after $t_c + \delta$, where δ exceeds the oracle latency Δ_o. Settlement is lazy: the outcome is fixed at commit time up to the oracle's future output, so no party holds an option after committing.

One-sided crowding fee. Let OI_j^L and OI_j^S be the aggregate long and short open interest (in units) across all accounts, and $\iota_j = (\mathrm{OI}_j^L - \mathrm{OI}_j^S)/(\mathrm{OI}_j^L + \mathrm{OI}_j^S) \in [-1, 1]$ the imbalance. Positions on the crowded side $\mathrm{sgn}(\iota_j)$ accrue a fee on notional at rate

$$r_j = \bar{r}_j \, \frac{\big(|\iota_j| - \theta_j\big)^+}{1 - \theta_j}. \tag{2}$$

The fee is debited and *never credited* to the other side.

Remark 4.6 (Why one-sided). A zero-sum funding transfer between simulated accounts, as on real perpetual venues, would give the receiving side a positive drift and break the supermartingale property on which Section 5 rests: a zero-skill bot could farm funding by holding the minority side. Debiting without crediting keeps every cost non-negative while still discouraging the herding that creates correlated liabilities (Section 7.8).

4.5 Risk checks

A fill that increases risk, that is, opens or enlarges a position, is admitted only if after the fill (i) $\mathrm{Lev}_a \leq \lambda$, (ii) the account holds at most n_{pos} open markets, and (iii) for funded accounts, the stress-coverage invariant (Invariant I1) holds. Risk-reducing fills are always admitted. A rejected fill leaves the account unchanged and is not charged. For a market order, checks (i) and (ii) depend only on the account and on reports published within Δ_{sync} of its fill observation. Check (iii) depends on global state that other accounts, possibly controlled by the same owner, can change before settlement. It is therefore applied when the order is committed, and an admitted order reserves its stress headroom until it settles, so that lazy settlement does not give the owner an option to have an unfavorable fill rejected.

4.6 Breach, pass and claims

Breach. An account in EVAL_k or FUNDED fails when its equity, evaluated on a synchronized set of admissible observations, is at or below $\max(f_a, f_a^{\mathrm{day}})$. Anyone may prove a breach (Algorithm 2) and receives a bounty b_{br} from the account's escrow. The synchronization bound Δ_{sync} prevents a prover from combining prices from different moments to fabricate a loss on a hedged book.

Pass. An account in EVAL_k that is flat (no positions and no pending orders) with $B_a \geq S(1+T_k)$ may call PASS. If $k < K$ it moves to EVAL_{k+1} with $B_a \leftarrow S$. If $k = K$ it becomes FUNDED with $B_a \leftarrow S_f$ when the capacity condition of Section 7.7 holds, and QUEUED otherwise.

Claims. A flat funded account with $G := B_a - S_f \geq g_{\min} S_f$ may call CLAIM (Algorithm 3). The owner receives αG USDC from the payout reserve, the balance resets to $B_a \leftarrow S_f$, and

Table 1: Transaction types and their effect on the global state.

Transaction	Caller	Preconditions	Effect
BUY(R, *ref*)	anyone	$R \in \mathcal{R}_k$; fee $F_k(R)$ paid	mint account in EVAL_1 with $B_a = S$; route the fee through the waterfall (Definition 6.1)
COMMIT(o)	owner or session key	account active; no pending market order; $j \in \mathcal{M}_{R_a}$	record o with commit time t_c
SETTLE($\mathsf{o}, \omega^\star, \Omega$)	anyone	Algorithm 1	record the fill or the rejection; keeper reward
TRIGGER($\mathsf{o}, \omega, \Omega$)	anyone	conditional order executable with a fresh ω	record the fill; keeper reward
CANCEL(o)	owner	conditional order	effective δ after the cancel is committed
PROVEBREACH(a, Ω)	anyone	Algorithm 2	FAILED; bounty b_{br}
PASS(a)	owner	flat; $B_a \geq S(1+T_k)$	next phase, FUNDED or QUEUED
CLAIM(a)	owner	Algorithm 3	αG USDC to the owner; balance reset
RETIRE(a)	owner	funded, flat and $B_a > \max(f_a, f_a^{\text{day}})$	CLOSED; escrow released to V_{res}
FUNDNEXT()	anyone	queue non-empty; capacity available	admit the head of $\mathcal{Q}$ to FUNDED
EPOCH()	anyone	epoch elapsed	controller update (Algorithm 4)

$W_a \leftarrow W_a + G$. The floor $f_a = S_f(1 - D_f)$ is unchanged, so every claim cycle starts from the same distance to failure. Claims above an instant threshold x_{inst} are released after a window Δ_{ch} during which they can be challenged *only* with objective evidence: an oracle incident, meaning a deviation beyond ε_x between primary and secondary reports affecting a fill of the account, or a proof that a protocol invariant was violated. Trading style can never be challenged.

4.7 Global state and transition function

Following [63], we write the global state as

$$\mathcal{S} = (\mathcal{A},\ \mathcal{O},\ \mathcal{V},\ \Theta,\ \mathcal{Z},\ \mathcal{Q}),$$

comprising accounts, pending orders, vault balances $\mathcal{V} = (\text{V}_{\text{res}}, \text{V}_{\text{ins}}, \text{V}_{\text{sus}}, \text{V}_{\text{dev}}, \text{V}_{\text{burn}})$ together with escrows, parameters Θ, controller state $\mathcal{Z}$ (Section 7) and the funding queue $\mathcal{Q}$. A transaction T maps the state by $\mathcal{S}' = \Upsilon(\mathcal{S}, \mathsf{T})$, where Υ is the identity whenever a precondition fails. Table 1 lists the transaction types, and Algorithms 1 to 3 specify the three that move value.

5 Zero-skill valuation

This section is the core of the paper. We show that fair execution turns every zero-skill strategy into a supermartingale, derive closed-form bounds on what such strategies can extract, and turn the bounds into prices and liabilities.

5.1 Fair execution implies no free edge

Assumption 5.1 (Fair-execution environment)**.**

(A1) Oracle prices satisfy Assumptions 2.3 and 2.4.

(A2) The strategy is zero-skill (Definition 2.5).

Algorithm 1 SETTLE($\mathsf{o}, \omega^\star, \Omega$): deterministic settlement of a market order

Input: pending $\mathsf{o} = (a, j, \Delta q, \mathsf{MKT}, \cdot, t_c)$; report $\omega^\star = (j, \hat{P}, c, u^\star, u^{\star-})$; reports Ω for the other open markets of a

require $\mathsf{Adm}(\omega^\star)$ and $u^{\star-} < t_c + \delta \le u^\star$ ▷ first eligible report
require every $\omega \in \Omega$ admissible and $\max_{\omega\in\Omega} |u_\omega - u^\star| \le \Delta_{\text{sync}}$
accrue the crowding fees of a up to $u^\star$
if $E_a \le \max(f_a, f_a^{\text{day}})$ at the prices of $\omega^\star$ and Ω **then**
    fail a as in lines 5–6 of Algorithm 2; **return**
end if
$P^{\text{fill}} \leftarrow \hat{P}\,(1 + \operatorname{sgn}(\Delta q)\, s_j(\Delta q, \omega^\star))$
$a' \leftarrow a$ with position j updated by $(\Delta q, P^{\text{fill}})$, realized PnL booked and fee $\varphi_j\,|\Delta q|\,P^{\text{fill}}$ debited
if o increases risk **and** a check of Section 4.5 fails for a' **then**
    mark o rejected ▷ no state change, no fee
else
    $a \leftarrow a'$
end if
$t_a \leftarrow u^\star$; pay the execution reward to the sender from esc_a

Algorithm 2 PROVEBREACH(a, Ω): permissionless enforcement

Input: $\text{ph}_a \in \{\text{EVAL}_k, \text{FUNDED}\}$; reports Ω covering every open market of a

require every $\omega \in \Omega$ admissible, $u_\omega > t_a$, and $\max_\omega u_\omega - \min_\omega u_\omega \le \Delta_{\text{sync}}$
settle any pending market order of a whose fill time precedes $\min_\omega u_\omega$
compute E_a at the prices of Ω, including accrued crowding fees
require $E_a \le \max(f_a, f_a^{\text{day}})$
close all positions at the prices of Ω; $\text{ph}_a \leftarrow$ FAILED
pay b_{br} from esc_a to the sender; release the remaining escrow to V_{res}

(A3) $\delta \ge \Delta_o$ (on chain, $\delta \ge \Delta_o + 1\,\text{s}$; Section 2); market orders fill at $u^\star(\mathsf{o})$ (Definition 4.4) and cannot be canceled; cancellations of conditional orders take effect after δ.

(A4) Spreads, fees and crowding charges are non-negative and never credited.

(A5) At every report, the leverage of a live account is at most λ, and relative price moves between consecutive admissible reports are bounded by $\bar{\gamma}$, with $\lambda\bar{\gamma} < 1$.

(A6) Conditional orders satisfy Assumption 4.5, and $s_{0,j} \ge v_j(\Delta_f)$, the selection value of Proposition 5.3 or, for non-Brownian prices, a bound on the expected relative improvement from selecting a report within a window of length Δ_f.

Condition (A5) is a modeling device that keeps equity positive; price moves beyond $\bar{\gamma}$ are the insurance fund's responsibility (Section 6). The engine enforces the leverage cap at risk-increasing fills (Section 4.5); adverse moves can raise leverage above λ between fills, so λ in (A5) stands for the largest leverage that a live account actually reaches.

Lemma 5.2 (Information alignment)**.** *Under (A2) and (A3), every position change recorded at a fill time $u^\star(\mathsf{o})$ is $\mathcal{G}_{u^\star(\mathsf{o})}$-measurable.*

Proof. The order is $\mathcal{F}_{t_c}$-measurable by (A2), and $\mathcal{F}_{t_c} = \mathcal{G}_{t_c+\Delta_o} \subseteq \mathcal{G}_{u^\star}$ because $u^\star \ge t_c + \delta \ge t_c + \Delta_o$. □

Proposition 5.3 (Selection value)**.** *Suppose the price of market j is a driftless geometric Brownian motion with volatility σ_j, as under Assumption 2.4, and an executor chooses, with hindsight,*

Algorithm 3 CLAIM(a): payout of a funded account

Input: $\mathrm{ph}_a = \mathrm{FUNDED}$; the sender is the owner; a is flat with no pending orders

$G \leftarrow B_a - S_f$; **require** $G \geq g_{\min} S_f$
$B_a \leftarrow S_f$; $\quad W_a \leftarrow W_a + G$
$x \leftarrow \alpha G$ ▷ in USDC
if this is the first claim of a **and** $\rho > 0$ **then**
    $x \leftarrow x + \rho F_a$ ▷ fee refund
end if
if $x \leq x_{\mathrm{inst}}$ **then**
    transfer x from $\mathrm{V}_{\mathrm{res}}$ to the owner
else
    lock x in $\mathrm{V}_{\mathrm{res}}$ as a pending claim, released at $t + \Delta_{\mathrm{ch}}$ unless challenged with objective evidence
end if

the most favorable report within a window of length w. The expected relative price improvement over the report at the start of the window is at most

$$v_j(w) = 2e^{\sigma_j^2 w/2}\,\Phi(\sigma_j\sqrt{w}) - 1 = \sigma_j\sqrt{2w/\pi} + O(\sigma_j^2 w),$$

where Φ is the standard normal distribution function, and the leading term is attained.

The bound $v_j(w)$ holds for buyers and sellers alike. It is the exact value for a seller if the log-price, rather than the price, is a driftless Brownian motion; under the geometric Brownian motion of Proposition 5.3, the seller's exact value is smaller by $\sigma_j^2 w/4 + O((\sigma_j^2 w)^{3/2})$ (Appendix C).

Proof. See Appendix C. The leading term is $\sigma_j \mathbb{E}[\max_{s \leq w} B_s] = \sigma_j \mathbb{E}\,|B_w|$ by the reflection principle. □

For a market with 60% annualized volatility and $\Delta_f = 2\,\mathrm{s}$, $v_j(\Delta_f) \approx 1.2$ basis points, so condition (A6) costs traders almost nothing while closing the last channel through which an executor could select a favorable price.

Theorem 5.4 (No free edge)**.** *Under Assumption 5.1, let $u_0 < u_1 < \cdots$ be the successive publish times of admissible reports of the markets of R_a, at which the account is marked to market. The evaluation PnL $X_n = E_a(u_n) - S$ and the funded process $Y_n = W_a(u_n) + E_a(u_n) - S_f$ are $(\mathcal{G}_{u_n})$-supermartingales. For an account that uses conditional orders, the proof establishes this with the qualification stated after it.*

Proof. Between u_n and u_{n+1} the position vector q_n is constant and, by Lemma 5.2, $\mathcal{G}_{u_n}$-measurable. The mark-to-market increment $q_n \cdot (P(u_{n+1}) - P(u_n))$ therefore has zero $\mathcal{G}_{u_n}$-conditional mean by Assumption 2.4; optional sampling applies between the stopping times u_n and u_{n+1} because each oracle price changes only at its own reports, by a bounded factor (A5). Spreads, fees and crowding charges reduce equity by non-negative amounts (A4). A conditional order becomes executable at the first report at or after $t_c + \delta$ that satisfies its trigger, a $\mathbb{G}$-stopping time at which, by the argument of Lemma 5.2, a fill would be $\mathbb{G}$-measurable. Relative to a fill at that report, the executor's choice of report within the freshness window improves the fill by at most $v_j(\Delta_f)$ per unit of notional in conditional expectation, which the base half-spread offsets (A6); for the supermartingale inequality, a conditional fill is attributed to that first executable report. A claim moves G from E_a to W_a and leaves Y unchanged. Hence X and Y are local supermartingales. By (A5) equity stays positive, so both processes are bounded below, and Fatou's lemma makes them true supermartingales. □

Remark 5.5. Theorem 5.4 is the formal content of the statement that the simulator does not leak. It identifies exactly which design choices are load-bearing: the commit delay must exceed oracle latency, market fills must be first-report deterministic, nothing may be credited, and the base spread must dominate the residual selection value. Each corresponds to a protocol invariant in Section 12.2.

The conditional-order step of the proof is rigorous within the price model of Proposition 5.3. Let $\tilde{u}$ be the publish time of the first executable report. Whatever the executor knows, Assumption 4.5 confines the fill to a report in $[\tilde{u}, \tilde{u}+\Delta_f]$, and that fill differs from the attributed one by a price improvement no larger than the hindsight maximum over the window; by the strong Markov property, its conditional expectation given $\mathcal{G}_{\tilde{u}}$ is at most $v_j(\Delta_f)$ per unit of notional. Two gaps remain. For non-Brownian prices, the bound in (A6) is assumed rather than derived. At report times between $\tilde{u}$ and the actual fill, the recorded equity differs from the attributed equity by an amount that depends on the executor's later choice, so at those times the supermartingale property is established for the attributed equity, not for the recorded one. Both are open questions (Section 12.3), and the results below that rest on Theorem 5.4 inherit them for accounts that use conditional orders.

5.2 The pass-probability bound

Fix a phase k with target T_k and maximum loss D_k. The account is checked for breaches at a sequence of $\mathbb{G}$-stopping times $c_1 < c_2 < \cdots$ among the report times, with $c_n \to \infty$, and fails at the first check at which $E_a \le \max(f_a, f_a^{\text{day}})$. The engine guarantees such a check only at the fill report of each settlement (Algorithm 1). A breach that a keeper finds at another report u is enforced only if its proof lands before any settlement with a later fill report, since proofs need reports published after the last settlement (Algorithm 2); otherwise a breach from which the account has recovered is erased, and whether the race is lost can depend on prices after u. We therefore assume that monitoring keepers win this race, so that every breach found at a check is enforced; the bounds below and the results built on them rest on this assumption, which the engine does not yet guarantee (Section 12.3). Let $\xi_k \ge 0$ bound the expected overshoot $f_a - E_a$ below the static floor at failure, in units of S, conditionally on failure by any given check, and write $D'_k = D_k + \xi_k$ for the *effective maximum loss*; Proposition 5.14 bounds ξ_k.

Theorem 5.6 (Pass-probability bound)**.** *Under Assumption 5.1, every zero-skill strategy passes phase k with probability at most*

$$\bar{p}_k = \frac{D'_k}{T_k + D'_k}.$$

For an evaluation with K phases, the probability of passing all phases is at most $\bar{p}(R) = \prod_{k=1}^{K} \bar{p}_k$.

Proof. Let τ be the phase's termination time (pass, failure, or abandonment, after which X is constant) and fix $n \in \mathbb{N}$. Partition the sample space into the events A_n (passed by c_n), F_n (failed by c_n) and N_n (neither). On A_n the account is flat with $B_a \ge S(1+T_k)$, so $X_{\tau\wedge c_n} \ge T_k S$. On N_n, the check at c_n found $E_a > f_a$, so $X_{\tau\wedge c_n} > -D_k S$. On F_n, $\mathbb{E}[X_{\tau\wedge c_n} \mid F_n] \ge -(D_k + \xi_k)S$ by the definition of ξ_k. Optional stopping for the supermartingale X holds at the bounded stopping times $\tau \wedge c_n \wedge u_m$, and it extends to $\tau \wedge c_n$ by Fatou's lemma as $m \to \infty$, because $X + S \ge 0$ by (A5). Hence

$$0 = X_0 \ge \mathbb{E}[X_{\tau\wedge c_n}] \ge T_k S\,\mathbb{P}(A_n) - D'_k S\,(\mathbb{P}(F_n) + \mathbb{P}(N_n)) = T_k S\,\mathbb{P}(A_n) - D'_k S\,(1 - \mathbb{P}(A_n)),$$

so $\mathbb{P}(A_n) \le D'_k/(T_k + D'_k)$. Letting $n \to \infty$ proves the first claim. For several phases, apply the same argument conditionally on $\mathcal{G}$ at the start of each phase, where the balance is reset to S, and multiply. □

The bound is the gambler's-ruin probability [22], attained, up to trading costs and overshoot, by a driftless strategy that bets until it hits either barrier. It holds for *every* zero-skill strategy, with any position sizing, any number of trades and any stopping rule.

5.3 The lifetime payout bound

Theorem 5.7 (Lifetime payout bound)**.** *Under Assumption 5.1, the expected cumulative gross claims of a funded account under any zero-skill strategy satisfy* $\mathbb{E}[W_\tau] \le D'_f S_f$*, where* $D'_f = D_f + \xi_f$*. The owner's expected lifetime payout is therefore at most*

$$V_f(R) = \alpha\, D'_f\, S_f.$$

Proof. By Theorem 5.4, $Y = W + E - S_f$ is a supermartingale with $Y_0 = 0$. Let τ be the time of failure or retirement and c_n the checks, as in Theorem 5.6. For each n, optional stopping at $\tau \wedge c_n$, justified as in the proof of Theorem 5.6 because $Y + S_f = W + E \ge 0$, gives $\mathbb{E}[W_{\tau\wedge c_n}] \le \mathbb{E}[S_f - E_{\tau\wedge c_n}]$. At failure, $\mathbb{E}[S_f - E \mid \text{failed}] \le D'_f S_f$; otherwise $S_f - E < D_f S_f$, either at retirement, which requires equity above the floors (Table 1), or at the check c_n. Hence $\mathbb{E}[W_{\tau\wedge c_n}] \le D'_f S_f$ for all n. Since W is non-decreasing and $c_n \to \infty$, monotone convergence gives $\mathbb{E}[W_\tau] \le D'_f S_f$, and payouts equal αW_τ. □

Remark 5.8 (Interpretation)*.* A funded account behaves like a reserve of $D'_f S_f$ virtual dollars that a zero-skill trader can at best convert into claims one for one before it is exhausted. The profit split converts virtual dollars into real ones at rate α. Claim frequency, minimum claims and the trader's choice of when to claim affect the timing of payouts but not their expected total.

5.4 Zero-skill value and the farm-proof price

Corollary 5.9 (Zero-skill value)**.** *Under Assumption 5.1, the expected USDC paid as profit share on an evaluation bought under* R*, by any zero-skill strategy, is at most*

$$V^\star(R) = \bar{p}(R)\, V_f(R) = \left(\prod_{k=1}^{K} \frac{D'_k}{T_k + D'_k} \right) \alpha\, D'_f\, S_f. \tag{3}$$

Proof. Combine Theorems 5.6 and 5.7 by the tower property. Queuing does not affect the bound because queued accounts do not trade. □

Let r_{ref} be the referral share of the fee, ρ the refund fraction, and $c_{\text{op}} \ge e$ the per-account operating cost: the escrow e taken from the fee (Definition 6.1), which pays keeper rewards and bounties and which a trader who runs her own keepers can recover in full, plus oracle fees and other costs not covered by escrow.

Definition 5.10 (Farm-proof price)**.** The *farm-proof price* of a rule set R is

$$F^\star(R) = \frac{V^\star(R) + c_{\text{op}}}{1 - r_{\text{ref}} - \rho\, \bar{p}(R)}, \tag{4}$$

which is defined whenever $r_{\text{ref}} + \rho\, \bar{p}(R) < 1$.

At a price $F \ge F^\star(R)$, the protocol's expected net revenue from a zero-skill buyer is at least $(1 - r_{\text{ref}})F - \rho F \bar{p}(R) - V^\star(R) - c_{\text{op}} \ge 0$. Referral commissions appear in the denominator because self-referral is free in a permissionless system: a commission is economically a discount.

Example 5.11. Take $K = 1$, $S = S_f = \$25{,}000$, $T_1 = 10\%$, $D_1 = D_f = 5\%$ and $\alpha = 0.8$, and ignore overshoot, costs, referrals and refunds. Then $\bar{p} = 5/15 = 1/3$, $V_f = 0.8 \cdot 0.05 \cdot 25{,}000 = \$1{,}000$ and $F^\star = V^\star = \$333.33$. Halving both loss limits to 2.5% gives $\bar{p} = 0.2$, $V_f = \$500$ and $F^\star = \$100$. Further examples are tabulated in Appendix B.

Proposition 5.12 (What the price must and need not cover)**.** *Under Assumption 5.1, the following rules can only lower the expected payout of a zero-skill strategy, so $V^\star$ remains a valid bound without them: daily loss limits, minimum trading days, consistency rules, caps on the number of positions, trading costs, and claim minimums or frequency limits. The following raise the zero-skill value and must be priced through* (3) *and* (4)*: a larger funded size (scaling), refunds, larger loss limits, and anything that increases the overshoot ξ, such as checking loss limits on closed equity or only at the end of the day, or monitoring less often.*

Proof. Rules in the first group add termination events at which the loss is at most the static limit, or reduce equity by non-negative amounts; the proofs of Theorems 5.6 and 5.7 go through unchanged. Rules in the second group enter the bound through S_f, ρ, D_k, D_f or ξ. □

5.5 Coalitions and Sybil strategies

The best-known attack on prop firms is multi-account hedging: buy two evaluations, go long on one and short on the other, and collect from whichever passes. In a permissionless system the attacker can run any number of accounts with arbitrarily correlated strategies.

Theorem 5.13 (Coalition bound)**.** *Let a coalition control accounts $a_1, \dots, a_n$, bought under rule sets $R_1, \dots, R_n$ at prices $F_1, \dots, F_n$ at arbitrary times, all possibly chosen adaptively, and trade them with any jointly determined zero-skill strategy. Under Assumption 5.1, the coalition's expected profit is at most*

$$\mathbb{E}\Big[\sum_{i=1}^{n}\Big(V^\star(R_i) + e - (1 - r_{\mathrm{ref}} - \rho\,\bar{p}(R_i))F_i\Big)\Big],$$

counting self-referral commissions, refunds, and keeper rewards and bounties paid from the coalition's own escrows as income. The bound is non-positive whenever $F_i \geq F^\star(R_i)$ for all i, because $c_{\mathrm{op}} \geq e$.

Proof. The coalition's information is public information, so each account, viewed in isolation, is traded by a zero-skill strategy whose orders may depend on the states of the other accounts. Theorems 5.4, 5.6 and 5.7 therefore apply to each account separately from its purchase onward and give $\mathbb{E}[\text{payout}_i \mid \text{purchase}] \leq V^\star(R_i)$, conditionally on the information at the purchase of a_i. Account a_i earns its refund ρF_i only after passing, with conditional probability at most $\bar{p}(R_i)$, and its escrow pays out at most e. Taking expectations and using linearity of expectation sums the bounds, whatever the dependence between accounts. □

Consequence: Sybil resistance by pricing

Correlation between accounts changes the *variance* of a coalition's outcome, not its *expectation*. A hedged pair turns a lottery into a near-certain result with the same mean. Multi-account hedging can only be profitable when evaluations are priced below their zero-skill value, so a protocol that prices at or above $F^\star$ needs no identity layer to stay solvent. Identity becomes a tool for offering *discounts* safely (Section 8), not a precondition for survival.

5.6 Monitoring and overshoot

Proposition 5.14 (Overshoot)**.**

(a) Under (A5), if the account is checked at every report, the equity at failure lies below the static floor by less than $\lambda\bar{\gamma}(1-D)S$, so $\xi \leq \lambda\bar{\gamma}(1-D)$.

(b) If log-prices are Brownian with volatility σ and the account is checked every Δ_m, then for small Δ_m the expected overshoot at failure of a single-market account at leverage λ is approximately $\xi \approx \lambda(1-D)\,\beta_{\mathrm{BGK}}\,\sigma\sqrt{\Delta_m}$, with $\beta_{\mathrm{BGK}} = -\zeta(1/2)/\sqrt{2\pi} \approx 0.5826$, where ζ is the Riemann zeta function.

Proof. (a) At the last check before failure, $E > f = S(1-D)$. Over one step, apart from crowding fees accrued during the step, equity falls by at most $N\bar{\gamma} \le \lambda E\bar{\gamma}$, so $E_{\text{fail}} \ge E(1-\lambda\bar{\gamma})$ and the overshoot satisfies $f - E_{\text{fail}} \le f - E(1-\lambda\bar{\gamma}) < \lambda\bar{\gamma}f$. (b) For a discretely monitored Brownian barrier, the expected overshoot of the log-price beyond the barrier, given that it is crossed, is $\beta_{\text{BGK}}\sigma\sqrt{\Delta_m} + o(\sqrt{\Delta_m})$ [6]; equity at the floor is $(1-D)S$, carrying notional $\lambda(1-D)S$. □

With $\lambda = 5$, $D = 5\%$, $\sigma = 60\%$ per year and $\Delta_m = 2\,\text{s}$, part (b) gives $\xi \approx 0.04\%$ of S. Overshoot is immaterial when limits are checked within seconds, but it becomes material when they are checked rarely, which is why Proposition 5.12 lists end-of-day checking among the features that must be priced.

5.7 Liability valuation

Proposition 5.15 (Remaining zero-skill liability)**.** *Under Assumption 5.1, at any time t:*

(a) for a funded account with equity $E_a(t)$, the expected remaining payout is at most

$$\alpha(E_a(t) - S_f(1-D'_f));$$

(b) for an account in $EVAL_k$ with equity $E_a(t)$, the probability of passing the current phase is at most

$$\pi_a(t) = \min\Big(1,\ \frac{E_a(t) - S(1-D'_k)}{(T_k + D'_k)\,S}\Big),$$

and its expected eventual payout is at most $\pi_a(t)\prod_{k'>k}\bar{p}_{k'}\,V_f(R_a)$;

(c) for a queued account, the expected eventual payout is at most $V_f(R_a)$.

Proof. Apply the arguments of Theorems 5.6 and 5.7 from time t, with distance $E_a(t) - S_f(1-D'_f)$ to failure in (a), and distances $E_a(t) - S(1-D'_k)$ to failure and $S(1+T_k) - E_a(t)$ to the target in (b). □

Definition 5.16 (Zero-skill and adjusted liability)**.** The *zero-skill liability* $\mathcal{L}^0(t) = \sum_{a\in\mathcal{A}}\mathcal{L}_a(t)$ sums the bounds of Proposition 5.15 over live accounts; when $\rho > 0$, each account's bound is increased by its outstanding refund ρF_a, weighted by $\pi_a(t)\prod_{k'>k}\bar{p}_{k'}$ in EVAL_k and by one once it has passed but not yet claimed. The *adjusted liability* is $\hat{\mathcal{L}}(t) = \hat{\psi}_t\,\mathcal{L}^0(t)$, where $\hat{\psi}_t \ge 1$ is a conservative estimate of the ratio between realized payouts and their zero-skill predictions on closed accounts (Section 7.1).

$\mathcal{L}^0$ is a sum of piecewise-linear functions of account equities that the engine can maintain incrementally, so the protocol's expected liability can be read at every block. Liability is recognized as it accrues: an evaluation approaching its target already carries almost the full value V_f, so passing does not make $\mathcal{L}^0$ jump. A conventional firm can estimate this quantity internally; Fairgame publishes it.

Summary of the valuation layer

Zero-skill pass probability	$\bar{p}_k = D'_k/(T_k + D'_k)$
Zero-skill funded value	$V_f = \alpha D'_f S_f$
Zero-skill evaluation value	$V^\star = (\prod_k \bar{p}_k)V_f$
Farm-proof price	$F^\star = (V^\star + c_{\text{op}})/(1 - r_{\text{ref}} - \rho\bar{p})$
Liability of a funded account	$\alpha(E_a - S_f(1-D'_f))$
Spread floor (selection value)	$v_j(w) \approx \sigma_j\sqrt{2w/\pi}$

6 Treasury and fee waterfall

6.1 Vaults

Fairgame holds USDC in five vaults and in per-account escrows:

- the **payout reserve** $\mathrm{V}_{\mathrm{res}}$, the only source of trader payouts; pending claims (Algorithm 3) are held within it, but in every coverage and stress condition, including the waterfall below and Section 7, $\mathrm{V}_{\mathrm{res}}$ denotes the reserve net of pending claims, since those amounts are already committed;
- the **insurance fund** $\mathrm{V}_{\mathrm{ins}}$, a capped backstop for events outside the model: oracle incidents, price moves beyond $\bar{\gamma}$ and contract defects;
- the **sustainability fund** $\mathrm{V}_{\mathrm{sus}}$, which pays recurring costs (oracle updates, keeper subsidies, gas sponsorship), holds the operating runway in conservative instruments and receives network fee monetization;
- the **development fund** $\mathrm{V}_{\mathrm{dev}}$, which pays contributors, audits and bug bounties;
- an optional **burn allocation** $\mathrm{V}_{\mathrm{burn}}$, used to buy back and burn S, in line with Sonic Labs' stated expectation that applications receiving its grants return value to the protocol layer, including through increased S burn [47].

6.2 The waterfall

Definition 6.1 (Fee waterfall)**.** A fee F paid in epoch k, with referral share r_{ref} (zero without a referrer) and escrow amount e, is allocated in order:

(1) *referral*: $r_{\mathrm{ref}}F$ to the referrer;

(2) *escrow*: e to esc_a;

(3) *reserve*: with $F_{\mathrm{net}} = (1 - r_{\mathrm{ref}})F - e$,

$$x_{\mathrm{res}} = \max\Big(w_{\mathrm{res}}^{\min} F_{\mathrm{net}},\ \min\left(F_{\mathrm{net}},\ \left(C^{\star}\widehat{\mathcal{L}} - \mathrm{V}_{\mathrm{res}}\right)^{+}\right)\Big);$$

(4) *insurance*: $x_{\mathrm{ins}} = \min\left(F_{\mathrm{net}} - x_{\mathrm{res}},\ w_{\mathrm{ins}}F_{\mathrm{net}}\right) \cdot \mathbf{1}\{\mathrm{V}_{\mathrm{ins}} < \mathrm{V}_{\mathrm{ins}}^{\star}\}$;

(5) *remainder*: $F_{\mathrm{rem}} = F_{\mathrm{net}} - x_{\mathrm{res}} - x_{\mathrm{ins}}$, split in proportions $(w_{\mathrm{sus}}, w_{\mathrm{dev}}, w_{\mathrm{burn}})$ that sum to one.

Proposition 6.2 (Waterfall properties)**.**

(i) The allocations are non-negative and sum to F, provided $e \le (1 - r_{\mathrm{ref}})F$ and $w_{\mathrm{res}}^{\min}, w_{\mathrm{ins}} \in [0, 1]$.

(ii) While the reserve is below target ($C < C^{\star}$), it receives $\max\left(w_{\mathrm{res}}^{\min}F_{\mathrm{net}}, \min(F_{\mathrm{net}}, C^{\star}\widehat{\mathcal{L}} - \mathrm{V}_{\mathrm{res}})\right)$, and the sustainability, development and burn allocations receive only what remains after the reserve and the insurance fund.

(iii) The reserve's allocation is non-increasing in $\mathrm{V}_{\mathrm{res}}$.

(iv) At least $w_{\mathrm{res}}^{\min}F_{\mathrm{net}}$ always reaches the reserve, so the reserve grows with the book even when coverage is met.

Proof. Immediate from Definition 6.1. □

6.3 Reserve sweep and insurance

When coverage exceeds an upper band, $C > C^{\max}$, at most a fraction w_{sw} of the excess $\mathrm{V}_{\text{res}} - C^{\max}\hat{\mathcal{L}}$ may be swept per epoch to the sustainability and burn allocations, and only as far as Invariant I1 permits. The insurance fund targets $\mathrm{V}^{\star}_{\text{ins}} = k_{\text{ins}} L^{\text{st}}$, a multiple of the stress liability of Section 7.8. It pays automatically if a due claim exceeds V_{res}, which Theorem 7.6 rules out while prices stay within the stress moves γ_j of their values at a time when stress coverage held, and otherwise only through timelocked governance for incident recovery.

6.4 Network revenue

Engine transactions are frequent and computationally non-trivial, so Fairgame generates substantial network fees. Under FeeM [52] the protocol currently receives 90% of the network fees for gas consumed in its registered contracts, credited to V_{sus}. Because the protocol sponsors traders' gas (Section 13.4), FeeM refunds 90% of the sponsorship spent in Fairgame's own contracts, but nothing spent in third-party contracts such as the oracle verifiers. FeeM pays in S, the asset in which sponsored gas is paid, so its rewards can fund sponsorship without conversion. The design does not depend on FeeM, which Sonic Labs may restructure [47]: the results of Sections 5 and 7 hold with FeeM revenue set to zero.

7 The self-balancing controller

The valuation layer prices zero skill. Real populations differ from that benchmark: most traders under-perform it, some genuinely out-perform it, and markets trend. The controller closes the loop between realized cash flows and the parameters that create liabilities. It runs once per epoch (one day in the reference parameterization) in the permissionless EPOCH transaction (Algorithm 4).

7.1 Signals

Let k index epochs. The controller observes:

- **coverage** $C_k = \mathrm{V}_{\text{res}}/\hat{\mathcal{L}}$, the reserve relative to the adjusted liability;
- the **loss ratio** $\ell_k = \bar{P}_k/\bar{F}_k$, the ratio of exponentially weighted moving averages of USDC paid in claims and of net fee revenue $(1 - r_{\text{ref}})F$;
- the **skill ratio** ψ_k, the exponentially weighted ratio, over accounts closed in recent epochs, of realized payouts to the zero-skill value $V^{\star}(R_a)$ at purchase, and its conservative version $\hat{\psi}_k = \max(1, \psi_k^{\text{UCB}})$ built from an upper confidence bound;
- **stress coverage** $C^{\text{st}}_k = \mathrm{V}_{\text{res}}/L^{\text{st}}$ (Section 7.8) and the **queue length** $Q_k = |\mathcal{Q}|$.

Remark 7.1 (The skill ratio as a leak detector). By Corollary 5.9, a population of zero-skill traders has $\psi \leq 1$ up to sampling error. A persistent $\psi > 1$ on a rule set means either that genuinely skilled traders are present or that Assumption 5.1 is violated somewhere, that is, that the simulator leaks. Both warrant attention; the second warrants it urgently. Fairgame publishes ψ per rule set and per market.

7.2 Levers and the price law

The controller acts through four levers: the price multiplier m_k, the offered menu $\mathcal{R}_k \subseteq \mathcal{R}$, funded capacity (Section 7.7) and, implicitly, open-interest headroom (Section 7.8). Prices are

$$F_k(R) = m_k\, F^{\star}(R) \quad \text{(anonymous)}, \qquad F^{v}_k(R) = \max\left(d_v m_k,\ m^{\min}_v\right) F^{\star}(R) \quad \text{(verified)}, \tag{5}$$

with the hard constraint $m_k \geq 1$, and a verified discount factor $d_v < 1$ floored at $m^{\min}_v$, available only within the per-identity caps of Section 8.

Algorithm 4 EPOCH(): controller update

Input: epoch k has elapsed
compute C_k, ℓ_k, $\widehat{\psi}_k$, C_k^{st}, Q_k and the elasticity estimate $\hat{\varepsilon}_{\nu,k}$
compute Δ_k by (7)
if $\hat{\varepsilon}_{\nu,k} \geq \bar{\varepsilon}$ **then** ▷ adverse selection: switch to quantity control
    $\Delta_k \leftarrow \min(\Delta_k, 0)$; $\bar{v}_{\max} \leftarrow (1-\chi_v)\,\bar{v}_{\max}$
end if
$z_{k+1} \leftarrow \Pi_{[z_{\min}, z_{\max}]}(z_k + \mathrm{clip}_{[-\chi,\chi]}(\Delta_k))$; $m_{k+1} \leftarrow e^{z_{k+1}}$
$\mathcal{R}_{k+1} \leftarrow \{R \in \mathcal{R} : V^\star(R) \leq \bar{v}_{k+1}\}$ with $\bar{v}_{k+1}$ from (8)
while $\mathcal{Q} \neq \emptyset$ **and** the capacity condition (9) holds for the head of $\mathcal{Q}$ **do**
    admit the head of $\mathcal{Q}$ to FUNDED
end while
if $C_k > C^{\max}$ **then**
    sweep at most $w_{\mathrm{sw}}(\mathrm{V}_{\mathrm{res}} - C^{\max}\widehat{\mathcal{L}})$ to $\mathrm{V}_{\mathrm{sus}}$ and $\mathrm{V}_{\mathrm{burn}}$, subject to Invariant I1
end if

7.3 Update law

Write $z_k = \ln m_k$. The controller applies a rate-limited, projected update

$$z_{k+1} = \Pi_{[z_{\min}, z_{\max}]}\Big(z_k + \mathrm{clip}_{[-\chi,\chi]}\left(\Delta_k\right)\Big), \tag{6}$$

$$\Delta_k = \eta_\ell(\ell_k - \ell^\star) + \eta_C\left(\frac{C^\star - C_k}{C^\star}\right)^+ - \eta'_C\left(\frac{C_k - C^{\max}}{C^{\max}}\right)^+ + \eta_Q\frac{Q_k}{Q_{\mathrm{ref}}}, \tag{7}$$

with $z_{\min} = 0$. The loss-ratio term is integral action toward the target $\ell^\star$; the coverage terms react to the balance sheet; the queue term raises prices when demand for funded capacity exceeds supply. The rate limit χ bounds the change in price per epoch by a factor e^χ, so traders face predictable prices.

7.4 Stability

To analyze the loss-ratio loop, consider a stationary population with *payout intensity*

$$\nu = \frac{\mathbb{E}[\text{payout per evaluation}]}{(1 - r_{\mathrm{ref}})\,F^\star},$$

so that $\ell = \nu/m$ at multiplier m. This idealization takes $\rho = 0$, so that payouts do not scale with the price, and assumes that ℓ_k responds to the current multiplier, ignoring the smoothing of ℓ_k and any lag between sales and claims. Zero-skill populations have $\nu \leq 1$ by construction of $F^\star$.

Proposition 7.2 (Local stability)**.** *Let ν be independent of m and $\nu > \ell^\star$. The unconstrained iteration $z_{k+1} = z_k + \eta_\ell(\nu e^{-z_k} - \ell^\star)$ has the unique fixed point $m^\star = \nu/\ell^\star$, which is locally asymptotically stable if $0 < \eta_\ell\ell^\star < 2$ and unstable if $\eta_\ell\ell^\star > 2$. Near the fixed point, convergence is monotone if $\eta_\ell\ell^\star < 1$ and oscillatory if $1 < \eta_\ell\ell^\star < 2$.*

Proof. The map $g(z) = z + \eta_\ell(\nu e^{-z} - \ell^\star)$ satisfies $g(z^\star) = z^\star$ for $z^\star = \ln(\nu/\ell^\star)$, and $g'(z^\star) = 1 - \eta_\ell\nu e^{-z^\star} = 1 - \eta_\ell\ell^\star$. The fixed point is locally asymptotically stable if $|g'(z^\star)| < 1$ and unstable if $|g'(z^\star)| > 1$; nearby iterates approach it monotonically if $g'(z^\star) > 0$ and with alternating sign if $g'(z^\star) < 0$. □

Proposition 7.3 (Global monotone convergence)**.** *Under the hypotheses of Proposition 7.2, with the coverage and queue terms of* (7) *inactive, suppose in addition that $\eta_\ell\,\nu\,e^{-z_{\min}} \leq 1$ and $z^\star \leq z_{\max}$. Then from every $z_0 \in [z_{\min}, z_{\max}]$ the iterates of* (6)*, with or without the rate limit, converge monotonically to $z^\star$. If instead $\nu \leq \ell^\star$, they converge to $z_{\min}$ and the multiplier rests at its floor.*

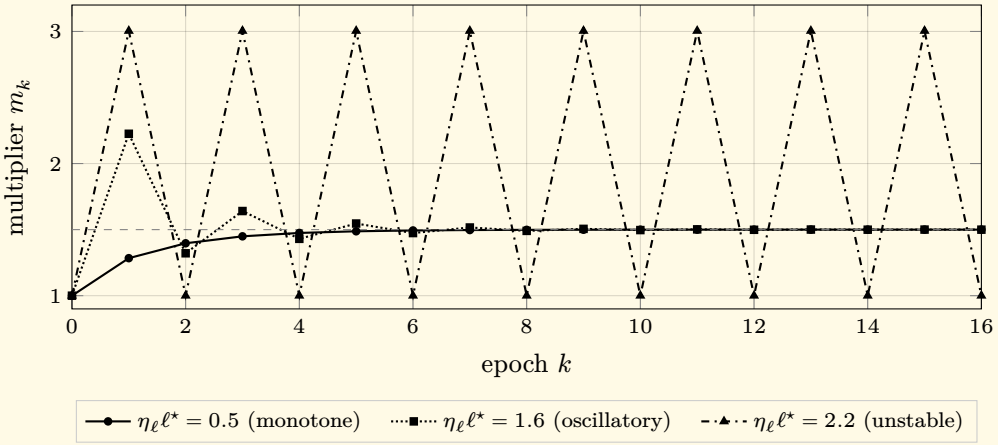


Figure 4: Iterates of the log-price update without rate limit, for payout intensity $\nu = 0.9$ and target loss ratio $\ell^\star = 0.6$ (fixed point $m^\star = 1.5$, dashed), starting at $m_0 = 1$ with bounds $[1, 4]$. Values are computed from the deterministic map of Proposition 7.2. Beyond $\eta_\ell \ell^\star = 2$ the fixed point is unstable, and here the iterates settle into a sustained oscillation between the floor $m = 1$ and $m \approx 3.0$; the rate limit χ damps such behavior but is no substitute for choosing gains inside the stability region.

Proof. See Appendix C. □

When $\nu \leq \ell^\star$, which is the typical case for retail populations that under-perform the zero-skill benchmark, prices sit at the farm-proof floor and the surplus $1-\ell$ of net revenue flows through the waterfall. Prices rise above the floor only when realized payouts demand it. Figure 4 illustrates the three regimes of Proposition 7.2. Lags shrink the stability region: if ℓ_k responds to m_{k-d} with a delay of d epochs, the linearized loop $x_{k+1} = x_k - \eta_\ell \ell^\star x_{k-d}$ is asymptotically stable if and only if $0 < \eta_\ell \ell^\star < 2\cos\left(d\pi/(2d+1)\right)$ [29], a bound that equals 1 for $d = 1$ and 0.62 for $d = 2$, so the reference gain of Table 6 tolerates a delay of up to two epochs.

7.5 Adverse selection

Raising prices changes *who* buys. If casual traders leave first, the remaining population is more skilled and ν rises with m. Let $\varepsilon_\nu = \mathrm{d}\ln\nu/\mathrm{d}\ln m$ be the elasticity of payout intensity with respect to price.

Proposition 7.4 (Adverse-selection condition)**.** *With $\ell(m) = \nu(m)/m$ and ε_ν evaluated at the fixed point, the fixed point of the loss-ratio loop is locally asymptotically stable if $0 < \eta_\ell \ell^\star(1-\varepsilon_\nu) < 2$ and unstable if $\eta_\ell \ell^\star(1-\varepsilon_\nu)$ lies outside $[0, 2]$. If $\varepsilon_\nu \geq 1$, raising the price does not lower the loss ratio, and price is not an effective lever.*

Proof. Since $\mathrm{d}\ell/\mathrm{d}z = \ell\,(\varepsilon_\nu - 1)$, the update map has $g'(z^\star) = 1 + \eta_\ell \ell^\star(\varepsilon_\nu - 1)$, and the criteria in the proof of Proposition 7.2 apply. □

The regime $\varepsilon_\nu \geq 1$ is the lemons spiral of insurance markets [1, 41]. The controller estimates $\hat{\varepsilon}_\nu$ by regressing $\ln\hat{\nu}_k$ on $\ln m_k$ over a rolling window. When $\hat{\varepsilon}_\nu \geq \bar{\varepsilon}$, it stops raising prices and switches to quantity controls (Algorithm 4): it tightens the menu, which reduces new liabilities directly rather than through demand, while admissions to FUNDED remain limited by the capacity condition (9).

7.6 The dynamic rule menu

Because $V^\star$ depends on the rules only through (3), rule sets with equal zero-skill value are economically interchangeable for the protocol, and tighter rules make evaluations cheaper (Figure 5).

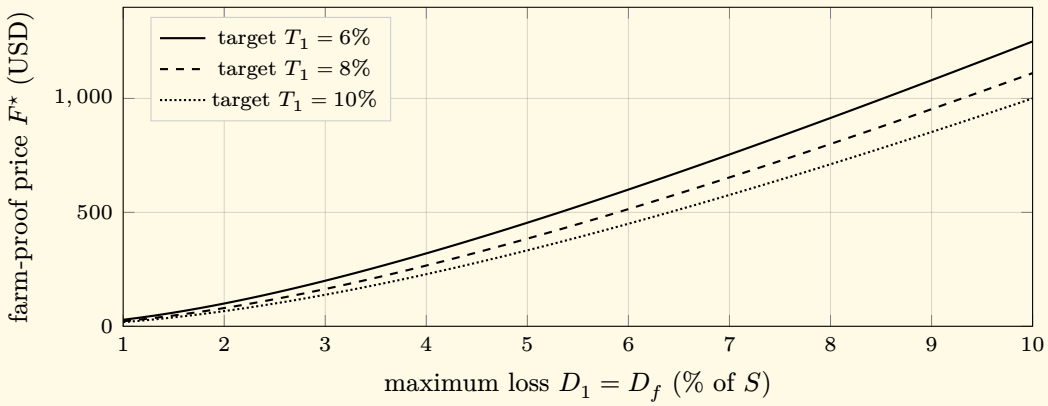


Figure 5: Farm-proof price of a one-phase evaluation with $S = S_f = \$25{,}000$ and $\alpha = 0.8$ as a function of the maximum loss, ignoring overshoot, costs, referrals and refunds: $F^\star = \alpha S D^2/(T_1 + D)$. The price is convex in the loss limit, so tightening the limit is the cheapest way to offer affordable evaluations.

The controller exposes this trade-off as a menu:

$$\mathcal{R}_k = \{R \in \mathcal{R} : \; V^\star(R) \le \bar{v}_k\}, \qquad \bar{v}_k = \bar{v}_{\max} \cdot \mathrm{clip}_{[v_{\text{floor}},1]}\Big(\frac{C_k - C^{\min}}{C^\star - C^{\min}}\Big). \tag{8}$$

When cash out outpaces cash in, coverage falls and the most generous rule sets (large accounts, wide loss limits) leave the menu, while tight rule sets remain on sale at unchanged prices. The rules of *existing* accounts never change. The split α of new rule sets can serve as an additional lever within $[\alpha_{\min}, \alpha_{\max}]$. Since $V^\star$ is linear in α, a change in split is economically equivalent to a proportional change in price, although traders perceive the two differently.

7.7 Capacity and the funding queue

Passing does not create zero-skill liability, because Proposition 5.15 recognizes it as an evaluation approaches its target. What admission to FUNDED creates is the ability to generate immediate USDC claims, that is, stress exposure. A passing account is therefore admitted only if

$$C_k \ge C^\star \qquad \text{and} \qquad \mathrm{V}_{\text{res}} \;\ge\; (1+\beta)\big(L^{\text{st}} + \rho F_a + \alpha\,\lambda\,\gamma_{\max}\,S_f\big), \tag{9}$$

where $\gamma_{\max} = \max_j \gamma_j$ and ρF_a is the passing account's outstanding refund. The second condition reserves enough stress headroom for that refund and for the new account to use its full leverage in any single market without being blocked by Invariant I1. Otherwise the account enters the FIFO queue $\mathcal{Q}$ and is admitted by a later EPOCH, or by any caller of FUNDNEXT, once (9) holds. A long queue is a public signal of scarce capacity and raises prices through η_Q in (7).

7.8 Stress coverage

Zero-skill pricing controls the *expected* liability. Correlated winners, such as many funded accounts long the same asset in a rally, create a *tail* liability that pricing does not address (Remark 2.6). Fairgame bounds it directly.

Definition 7.5 (Stress liability)**.** Let $\gamma_j > 0$ be a stress move for market j over the claim horizon, and let $\mathrm{OI}_j^{L,f}$ and $\mathrm{OI}_j^{S,f}$ be the aggregate long and short open interest of funded accounts. The stress liability is

$$L^{\text{st}}(t) = \Gamma(t) + \alpha \sum_{j \in \mathcal{M}} \gamma_j\, P_j(t)\,\big(\mathrm{OI}_j^{L,f}(t) + \mathrm{OI}_j^{S,f}(t)\big),$$

$$\Gamma(t) = \sum_{a \text{ funded}} \Big(\alpha\,\big(E_a(t) - S_f\big)^+ + \rho F_a \mathbf{1}_{\{W_a(t)=0\}}\Big),$$

where Γ is the value that funded accounts could claim immediately, including first-claim refunds not yet paid, and α is the largest profit split among funded accounts.

Invariant I1 (Stress coverage)**.** Every risk-increasing fill by a funded account, every admission to FUNDED and every reserve sweep leaves $\mathrm{V}_{\mathrm{res}} \geq (1+\beta)\, L^{\mathrm{st}}$; transitions that would violate this inequality are rejected.

Price moves can lower stress coverage between such transitions, so the inequality is enforced at the transitions that the protocol controls rather than in every state. Nothing in the protocol restores it once prices have eroded it: C_k^{st} is computed and published at every epoch (Algorithm 4) but enters no update rule, and no transition forces funded positions down. Until claims, closures, failures or inflows restore the inequality, Invariant I1 only blocks new risk, and Theorem 7.6 gives no guarantee, because its hypothesis fails. How eroded stress coverage should be restored is an open design question (Section 12.3). Claims only increase the slack $\mathrm{V}_{\mathrm{res}} - (1+\beta)L^{\mathrm{st}}$, because a claim lowers L^{st} by at least as much as it lowers $\mathrm{V}_{\mathrm{res}}$.

Theorem 7.6 (Shock solvency)**.** *Suppose* $\mathrm{V}_{\mathrm{res}}(t) \geq (1+\beta)L^{\mathrm{st}}(t)$*, no risk-increasing fills or reserve sweeps occur on* $[t,t']$*, and* $\left|P_j(s)/P_j(t) - 1\right| \leq \gamma_j$ *for all markets* j *and all* $s \in [t,t']$*. Then the total USDC that funded accounts can claim, closing positions at arbitrary times in* $[t,t']$*, is at most* $\mathrm{V}_{\mathrm{res}}(t)/(1+\beta)$*.*

Proof. An account closing at time s can claim at most $\alpha\left(E_a(s) - S_f\right)^+$, plus its refund ρF_a if it has not claimed before. Costs are non-negative, and positions can only be reduced during $[t,t']$, so each unit of the position $q_{a,j}$ held at t is closed or marked at a price within $\gamma_j P_j(t)$ of $P_j(t)$. Hence

$$E_a(s) \leq E_a(t) + \sum_j \left|q_{a,j}\right|\, P_j(t)\, \gamma_j.$$

Using $(x+y)^+ \leq (x)^+ + (y)^+$ and summing over funded accounts, total claims are at most $\Gamma(t) + \alpha \sum_j \gamma_j P_j(t) \sum_a \left|q_{a,j}\right| = L^{\mathrm{st}}(t) \leq \mathrm{V}_{\mathrm{res}}(t)/(1+\beta)$. Once an account has claimed, its balance is reset and it cannot reopen risk during $[t,t']$, so it generates no further claims. □

Remark 7.7. Gross rather than net open interest appears because accounts on opposite sides may close at different moments, each near its own favorable extreme. If all claims occurred at a single instant, $\max(\mathrm{OI}_j^{L,f}, \mathrm{OI}_j^{S,f})$ would suffice. Through the admission rule, Invariant I1 acts as a dynamic open-interest cap that tightens automatically as the reserve shrinks or prices rise, and the one-sided crowding fee (2) prices the herding that consumes it.

8 Identity and reputation

8.1 Optional proof of personhood

Identity is never required. A trader may attach a *unique-human attestation* to an address, for example a zero-knowledge proof of personhood or of a government credential. The attestation unlocks the verified price of (5) for at most n_v evaluations per epoch.

Proposition 8.1 (Bounded discount exposure)**.** *Under Assumption 5.1, a verified identity extracts in expectation at most* $n_v(1 - m_v^{\min}) \max_{R \in \mathcal{R}} F^\star(R)$ *per epoch from discounted evaluations, and at most* $H\, n_v(1 - m_v^{\min}) \max_R F^\star(R)$ *over a useful life of* H *epochs. If the market cost of a fraudulent attestation exceeds this amount, farming discounts with fake identities is unprofitable.*

Proof. By Theorem 5.13 and (4), the expected profit of a zero-skill buyer on one evaluation bought at $F^v(R) \geq m_v^{\min} F^\star(R)$ is at most $(1 - r_{\mathrm{ref}} - \rho\bar{p}(R))(F^\star(R) - F^v(R)) \leq (1 - m_v^{\min})F^\star(R)$. Multiply by the cap n_v and the horizon H. □

Discounts are how Fairgame competes with conventional firms on price for the honest majority, without reintroducing the multi-accounting exposure that those firms manage through identity checks.

8.2 The trader passport

Each address accumulates a soulbound *passport* that aggregates the on-chain history of its accounts: evaluations bought, passed and failed; claim cycles and payouts; maximum drawdowns; and a commitment to the daily equity series of every funded account. From the daily returns $r_1, \dots, r_n$ of an account, anyone can compute the per-period (daily, not annualized) Sharpe ratio $\widehat{SR}$, the skewness $\hat{\gamma}_3$, the Pearson kurtosis $\hat{\gamma}_4$ (not excess kurtosis, so $\hat{\gamma}_4 = 3$ for normal returns) and the *probabilistic Sharpe ratio* [4]

$$\mathrm{PSR}(SR^*) = \Phi\left(\frac{(\widehat{SR} - SR^*)\sqrt{n-1}}{\sqrt{1 - \hat{\gamma}_3\widehat{SR} + \frac{\hat{\gamma}_4 - 1}{4}\widehat{SR}^2}}\right), \tag{10}$$

the probability that the true Sharpe ratio exceeds a benchmark SR^*. Because Fairgame observes thousands of accounts, the best of them look skilled by chance. The passport therefore reports the *deflated* Sharpe ratio [5], $\mathrm{DSR} = \mathrm{PSR}(SR_0^*)$, with

$$SR_0^* = \sqrt{\mathrm{Var}[\widehat{SR}]}\left((1-\gamma_E)\,\Phi^{-1}(1-\tfrac{1}{N}) + \gamma_E\,\Phi^{-1}(1-\tfrac{1}{N\mathrm{e}})\right), \tag{11}$$

where N is the number of independent trials (accounts observed), $\mathrm{Var}[\widehat{SR}]$ the cross-sectional variance of their Sharpe ratios and $\gamma_E \approx 0.5772$ the Euler–Mascheroni constant.

8.3 Uses of reputation

- **Discounts.** Verified addresses whose passports exceed a threshold receive a larger discount $1 - d_v$, within the caps of Proposition 8.1.
- **Scaling.** After each successful claim cycle that meets drawdown criteria, S_f grows by a factor $g > 1$ up to $S_{\max}$. Each increase is a new liability $\alpha D'_f\,\Delta S_f$ (Theorem 5.7) and must satisfy the capacity condition (9).
- **Direct funding.** Verified addresses with $\mathrm{DSR} \geq q_{\mathrm{dir}}$ may buy a funded account without an evaluation. Its zero-skill value is V_f rather than $V^\star$, because the pass probability is one, so it is priced at m_k times the farm-proof price (4) with $\bar{p} = 1$, that is, at $m_k(V_f + c_{\mathrm{op}})/(1 - r_{\mathrm{ref}} - \rho)$, before reputation discounts, a price that is defined when $r_{\mathrm{ref}} + \rho < 1$.
- **Copy eligibility and governance weight** (Sections 10 and 11).

9 Tournaments

Tournaments are Fairgame's launch product. They generate trading activity, a leaderboard and a first funded cohort before open-ended evaluations exist, and they cannot create unfunded liabilities.

Definition 9.1 (Tournament)**.** A tournament $\mathcal{T} = (b, N_{\min}, N_{\max}, S_t, R_t, \mathsf{score}, \zeta, (\pi_r)_r, n_{\mathrm{aw}}, R_f)$ has buy-in b; between $N_{\min}$ and $N_{\max}$ entrants, each trading an svUSD account of size S_t under rule set R_t over a fixed window; a scoring function; a prize share ζ distributed by rank weights π_r with $\sum_r \pi_r = 1$; and n_{aw} funded accounts under rule set R_f, awarded to the top n_{aw} entrants.

Proposition 9.2 (Solvency by construction)**.** *If the realized number of entrants $N \in [N_{\min}, N_{\max}]$ satisfies*

$$\zeta + r_{\text{ref}} + \frac{e}{b} + \frac{n_{\text{aw}}\, V_f(R_f)}{Nb} \le 1, \tag{12}$$

which holds for every such N if it holds at $N = N_{\min}$, then all prizes are paid from buy-ins, the zero-skill value of every awarded funded account is transferred to V_{res} at settlement, and the residual flows through the waterfall. The tournament creates no zero-skill liability that is not pre-funded.

Proof. Buy-ins total Nb. Prizes total ζNb; referrals and escrow total at most $(r_{\text{ref}} + e/b)Nb$; and the funded awards carry zero-skill liability $n_{\text{aw}} V_f(R_f)$ by Theorem 5.7. Condition (12) states that these sum to at most Nb. □

Remark 9.3 (Coalitions in tournaments). Prize money is fixed, so a coalition that enters many times can only redistribute prizes among entrants. The protocol's prize outflow is deterministic and its funded awards are pre-funded. Multiple entries are a fairness question, handled by optional per-identity entry caps, not a solvency question.

Remark 9.4 (Risk-shifting and scoring). Rank-order prizes [28] reward variance [7]: ranking on raw return favors all-in bets that a few entrants win by luck. Fairgame scores return per unit of drawdown,

$$\mathsf{score}_a = \frac{E_a^{\text{end}} - S_t}{S_t \, \max(\mathrm{MDD}_a, d_0)},$$

where MDD_a is the maximum drawdown as a fraction of S_t and d_0 a floor. It disqualifies entrants who breach the loss limits of R_t and requires a minimum number of trading days. Funded awards are then more likely to reach traders whose results reflect skill.

Bootstrapping. Open evaluations launch once the reserve accumulated from tournaments supports the initial menu: $\mathrm{V}_{\text{res}} \ge C^\star \hat{\mathcal{L}}_0$, where $\hat{\mathcal{L}}_0$ is the adjusted liability implied by the planned initial capacity, and $\mathrm{V}_{\text{res}} \ge (1+\beta)L^{\text{st}}$ (Invariant I1) holds at the expected initial open interest.

10 The copy-capital vault

Simulated payouts make skilled traders a cost: the better they trade, the more the reserve pays. The copy-capital vault, planned for a later phase, reverses this by trading real capital alongside them.

10.1 Structure

Liquidity providers deposit USDC into an ERC-4626 vault [43], as may the reserve out of its excess above $C^{\max}$. An executor mirrors the positions of selected funded accounts on an external perpetual-futures venue, scaling trader i's simulated position by a *copy weight* h_i expressed in units of simulated notional. Withdrawals are processed in epochs, as asynchronous redemption requests in the sense of ERC-7540 [31], because depositors can observe the vault's exposures on-chain and would otherwise exit ahead of losses.

10.2 Selection

A funded account is eligible when its deflated Sharpe ratio (11) exceeds a threshold $1 - \alpha_{\text{sel}}$ over a minimum track record, its drawdowns respect the vault's limits, and its owner holds a verified passport. Copy weights are capped per trader and per market.

10.3 Economics

Consider one claim cycle in which trader i's simulated profit is X_i (in svUSD) with mean μ_i, and write $X_i^- = (-X_i)^+$. The reserve pays $\alpha\,(X_i)^+$; the vault earns $h_i X_i$ less execution costs $h_i\kappa_i$.

Proposition 10.1 (Hedge ratio)**.** *The combined expected result of the reserve and the vault on trader i is*

$$\mathbb{E}[\Pi_i] = (h_i - \alpha)\,\mu_i - \alpha\,\mathbb{E}[X_i^-] - h_i\kappa_i.$$

(a) For a trader without skill ($\mu_i \le 0$), copying never improves the expected result relative to not copying. It can reduce variance, as a partial hedge of the payout liability, but only at the expected cost $h_i(\kappa_i - \mu_i) \ge 0$.

(b) For a skilled trader, the result is positive whenever $(h_i - \alpha)\mu_i > \alpha\mathbb{E}[X_i^-] + h_i\kappa_i$. The weight $h_i = \alpha$ neutralizes the mean of the payout liability apart from the loss leg and costs, and with $h_i > \alpha$ the result increases with skill μ_i and turns positive once this condition holds.

Proof. Since $\mathbb{E}[(X_i)^+] = \mu_i + \mathbb{E}[X_i^-]$, we have $\mathbb{E}[h_i X_i - h_i\kappa_i - \alpha\,(X_i)^+] = (h_i - \alpha)\mu_i - \alpha\mathbb{E}[X_i^-] - h_i\kappa_i$. Relative to $h_i = 0$, copying changes the expected result by $h_i(\mu_i - \kappa_i)$, which is non-positive when $\mu_i \le 0$; this proves (a). Part (b) restates $\mathbb{E}[\Pi_i] > 0$. □

10.4 Execution synergy

The two-step engine gives the executor advance notice. A trader's COMMIT is public at t_c, while the simulated fill occurs at $u^\star \ge t_c + \delta$, so the executor can trade on the external venue within $[t_c, u^\star]$ and achieve fills roughly synchronous with the simulation. Commits are visible to everyone, so third parties can copy them too. This costs the simulation nothing and costs the vault only some execution quality.

10.5 Net-book hedging

Independently of selection, the reserve's exposure to correlated winners can be reduced by hedging a fraction of the funded book's aggregate delta on an external venue. Because the reserve pays only winners, its exposure is convex: economically, it is short a portfolio of call options on funded accounts. A linear hedge therefore reduces this exposure but cannot remove it. Fairgame treats net-book hedging as a risk-management overlay governed by the same timelocked process as other treasury actions.

11 Governance

11.1 Parameter classes

Every parameter belongs to one of the classes in Table 2. The guiding rule is that governance may tune the protocol but may not change a promise already made: rules of existing accounts, claims already earned and the immutable bounds of Table 3 are beyond its reach.

11.2 Voting power

Governance is decentralized progressively. At launch, slow-class changes are proposed by a multi-sig and executed through a timelock that anyone can monitor. In the target state, voting power is earned rather than bought: each verified passport holder receives non-transferable weight

$$v_i = \sqrt{P_i} \cdot \min\left(1,\ \tau_i/\tau_0\right), \tag{13}$$

where P_i is the cumulative USDC received in payouts and τ_i the funded tenure. The square root dampens concentration in the spirit of quadratic mechanisms [27], and the tenure factor resists

Table 2: Parameter classes.

Class	Changed by	Examples
Immutable	nobody	svUSD properties (S1)–(S3); immutability of an account's rules; no discretionary denial of claims; the hard bounds of Table 3; the invariants of Section 12.2
Slow	timelocked governance (at least 7 days)	catalog $\mathcal{R}$; market listings and oracle sources; waterfall weights; controller gains and targets within bounds; spreads and fees
Automatic	EPOCH	multiplier m_k; menu $\mathcal{R}_k$; queue admissions; reserve sweeps
Emergency	guardian	pause new evaluations and risk-increasing fills for at most Δ_{pause}; never pause claims of settled profits or move funds

Table 3: Hard bounds (proposed initial values).

Parameter	Bound	Rationale
Anonymous multiplier m_k	≥ 1	farm-proof pricing (Theorem 5.13)
Commit delay δ	$\geq$ 99.9th percentile of Δ_o plus 1 s	information alignment (Lemma 5.2)
Base half-spread $s_{0,j}$	$\geq v_j(\Delta_f)$	selection value (Proposition 5.3)
Profit split α	≤ 0.9	bounds V_f
Leverage cap λ	≤ 10	bounds overshoot (Proposition 5.14)
Stress buffer β	≥ 0.25	shock solvency (Theorem 7.6)
Coverage target $C^\star$	≥ 1.25	reserve adequacy
Rate limit χ	$\leq \ln 1.1$ per epoch	price predictability
Challenge window Δ_{ch}	≤ 72 h	timely payouts
Guardian pause Δ_{pause}	≤ 7 days	bounded emergency power

short-term capture. Because a concave weight rewards splitting payouts across addresses, voting weight is bound to a verified identity. Liquidity providers in the copy-capital vault vote on vault-specific parameters only. No governance action can violate an immutable rule; in particular, no vote can change the rules of an existing account or deny a claim.

11.3 Emergency powers

A guardian, initially a multisig and later elected under (13), may pause new evaluations and risk-increasing fills during an incident, for at most Δ_{pause}. It cannot pause claims of settled profits, alter balances or move funds, and every pause is logged on-chain with a stated reason.

12 Security analysis

12.1 Adversary model

The adversary controls any number of addresses, runs keepers, observes the mempool and all public state, and sees external-venue prices up to Δ_o ahead of the oracle. It cannot break standard cryptography, cannot obtain valid signatures on reports of its choosing from the primary and secondary oracle networks simultaneously, does not control validators holding a third or more of the stake, and does not hold a majority of governance weight. Table 4 summarizes the threats this adversary can mount and the mechanisms that address them.

Oracle trust. Since Pyth's upgrade of August 26, 2026, a primary report is valid when signed by three of five routers that Pyth describes as independently operated, in place of 13 of 19 Wormhole guardians; the signature scheme and the contract interface are unchanged, and the Pyth DAO upgraded the existing contracts in place [37, 39]. Trusting the primary therefore means trusting that no three of the five routers collude or are compromised, and that the Pyth DAO, which can

replace the verifying contract, does not install a malicious one. Secondary reports are signed by a Chainlink decentralized oracle network and checked by Chainlink's verifier contract [10]. Because Definition 2.2 requires an agreeing secondary report for every admissible observation, while fills and equity use the primary price, the second network bounds the damage from compromising the first rather than removing it. An adversary that controls Pyth's quorum alone can sign prices up to ε_x away from a genuine secondary report published within Δ_{sync}, and can choose the publish times on which first-report certificates rely; it can thus shift fills, and the equity used in breach proofs, by up to about ε_x of the notional involved. An adversary that controls the secondary network alone cannot move admissible prices, since fills and equity use the primary price, but it can withhold reports or sign disagreeing ones, which blocks admission and can mimic the oracle incidents accepted as evidence against claims (Section 4.6).

Oracle access and keeper liveness. Submitting a signed report on Sonic remains permissionless, but obtaining reports does not. Since the upgrade, every request to Pyth's Hermes service requires an API key, for which a free trial is included and paid plans cover ongoing use [37]; Chainlink's Data Streams API likewise serves reports only to authenticated accounts subscribed to the feed [11]. Every keeper therefore needs credentials from both providers. This raises the cost of entering the keeper market on which Assumption 4.5 relies, and it makes the providers' distribution services a common point of failure: an outage or loss of access stops all keepers at once, including the protocol backstop, which holds its own credentials paid from V_{sus}. During such an outage, market orders stay fixed at commit time and settle late but unchanged, provided the signed reports remain retrievable; conditional fills and breach proofs are delayed, so stop fills can slip and losses can overshoot floors by more than the monitoring interval assumed in Proposition 5.14 implies, a residual left to the insurance fund.

12.2 Protocol invariants

The following invariants, together with Invariant I1, are stated so that they can be checked mechanically: by property-based tests against the reference model and by formal verification of the deployed contracts.

Invariant I2 (Accounting)**.** The USDC held by protocol contracts other than the copy-capital vault equals the USDC balances of V_{res}, V_{ins}, V_{sus}, V_{dev} and V_{burn}, plus $\sum_a \mathsf{esc}_a$ and the buy-ins held for unsettled tournaments, with pending claims held within V_{res}.

Invariant I3 (Backed claims)**.** The total of pending claims never exceeds V_{res}.

In Invariants I2 and I3, V_{res} denotes the gross reserve, which includes the pending claims held within it; Invariant I3 thus states that the net reserve used in every coverage and stress condition (Section 6) is never negative.

Invariant I4 (Rule immutability)**.** The rule set R_a is constant over the life of every account a.

Invariant I5 (Non-transferability)**.** No transition changes B_a except fills, fees, crowding accruals, claims and phase resets of a itself.

Invariant I6 (Fair timing)**.** Every market fill uses a report with $u^- < t_c + \delta \le u$. Every conditional fill uses a report with $u \ge t_c + \delta$ and $t - u \le \Delta_f$. Every report applied to an account has $u > t_a$.

Invariant I7 (Non-negative costs)**.** No transition credits an account with a spread, fee or crowding amount.

Invariant I8 (Leverage)**.** After every admitted risk-increasing fill, $\mathrm{Lev}_a \le \lambda$ and the account holds at most n_{pos} open markets.

Table 4: Threat model.

Threat	Mitigation	Residual risk
Latency arbitrage: trading on information the oracle has not yet incorporated	commit delay $\delta \geq \Delta_o$ ($\Delta_o + 1$ s on chain); first-report fills; market orders cannot be canceled (Lemma 5.2)	none under Assumption 2.3
Selection of favorable reports by keepers	first-report certificate for market orders; freshness window and base spread for conditional orders (Proposition 5.3)	bounded by the spread
Cancellation option	market orders cannot be canceled; conditional cancels are delayed by δ	none
Oracle manipulation or failure	liquid markets only; primary report admitted only with an agreeing secondary report (3-of-5 Pyth routers and a Chainlink network); confidence and staleness bounds	one network: prices within ε_x, or delay; both: tail risk, insured
Price gaps through the floor	leverage cap; frequent permissionless monitoring; overshoot priced (Proposition 5.14)	tail risk, insured
Sybil farming and hedged coalitions	farm-proof pricing (Theorem 5.13); discounts only with attestation and caps (Proposition 8.1)	cost of fake identities
Correlated winners	stress invariant and dynamic OI caps (Theorem 7.6); crowding fee; capacity control	set by the choice of γ_j
Simulator leak through an unknown channel	skill-ratio monitoring per rule set and market; challenge window for large claims	detection lag
Keeper censorship or loss of oracle access	permissionless execution; bounties; protocol backstop keeper with its own Pyth and Chainlink credentials; deterministic lazy settlement	delay; slippage and overshoot beyond the priced level
MEV and front-running	outcomes fixed at commit time by future oracle output	none for the simulation
Contract defects	audits; invariant and differential testing; formal verification of value-moving logic; caps; bug bounty	residual, insured
Governance capture	immutable class; hard bounds; timelock; identity-bound, non-transferable votes	limited by hard bounds
Stablecoin freeze or de-peg	reserve diversification; insurance fund	residual

Invariant I9 (Bounds)**.** The parameters Θ satisfy the hard bounds of Table 3 at all times; in particular $m_k \geq 1$.

Invariant I10 (Deterministic payouts)**.** The amount of a claim is a function of the account's state alone, and an unchallenged claim is released no later than Δ_{ch} after it is made.

12.3 Open design questions

The engine leaves four points open. Each is stated where it arises, together with the assumption, if any, that the results there make about it.

(i) *Inadmissible first reports.* No outcome is specified for a market order whose certified first report is not admissible, and such an order blocks its account indefinitely (Section 4.3).

(ii) *Eroded stress coverage.* Price moves can push V_{res} below $(1+\beta)L^{\text{st}}$; the protocol then blocks new risk, but nothing actively restores the inequality, and Theorem 7.6 gives no guarantee until it holds again (Section 7.8).

(iii) *Erasure of breaches.* A breach found at a report other than a settlement's fill report can be erased by a later settlement before its proof lands; the pass and payout bounds assume that it is not (Section 5.2).

Table 5: Contract modules.

Module	Responsibility
`RuleSetRegistry`	catalog $\mathcal{R}$, rule-set hashes, menu flags for $\mathcal{R}_k$
`AccountNFT`	ERC-721 accounts locked to their owner (ERC-5192); phase and rule-set reference
`TradingEngine`	orders, deterministic settlement, positions, costs, breach proofs, passes and claims
`OracleAdapter`	verification of primary (Pyth) and secondary (Chainlink Data Streams) reports; first-report certificates; agreement, confidence and staleness checks
`RiskController`	liabilities $\mathcal{L}^0$ and L^{st}, signals, multiplier, menu, capacity and queue
`FeeRouter`	the waterfall of Definition 6.1
`Vaults`	payout reserve, insurance, sustainability, development and burn
`TournamentManager`	tournament lifecycle, scoring, prizes and funded awards
`IdentityGate`	attestation verification and per-identity caps
`Passport`	reputation statistics and daily-equity commitments
`Governance`	timelock and guardian; identity-bound reputation voting (later phase)
`CopyVault`	ERC-4626/ERC-7540 copy-capital vault and executor (later phase)

(iv) *Conditional orders.* The treatment of conditional fills in Theorem 5.4 is rigorous only within the selection-value model, and only for the attributed equity between the first executable report and the fill (Section 5.1); the bounds that rest on that theorem inherit this gap for accounts that use conditional orders.

13 Implementation on Sonic

13.1 Contract architecture

Table 5 lists the contract modules. Value-moving logic is concentrated in three of them (the trading engine, the fee router and the vaults), which keeps the surface for audits and formal verification small.

13.2 Oracles

Settling a market order requires a certificate that the submitted report is the first published at or after $t_c + \delta$. Pull oracles whose signed updates carry the previous publish time support this check directly: Pyth's EVM contract accepts an update in `parsePriceFeedUpdatesUnique` only if $\mathit{prevPublishTime} < \mathit{minPublishTime} \leq \mathit{publishTime}$ [36]. Since Pyth's upgrade of August 26, 2026, these updates are signed by a quorum of three of five independently operated routers instead of 13 of 19 Wormhole guardians, with the contract interface unchanged, and the upgraded contract is deployed on Sonic; keepers fetch updates from Pyth's Hermes service, which has required an API key since the upgrade [35, 39]. Chainlink Data Streams, whose verifier is also deployed on Sonic [12], serves as the secondary source for the agreement check of Definition 2.2. The per-market parameters $\kappa_{\max}$, ε_x, Δ_{sync} and Δ_f are set from measured feed behavior, and a market is listed only when both oracle networks provide robust feeds for it.

Several Sonic blocks can share one second, but block timestamps, like Pyth publish times, are whole seconds, so the timing conditions of Section 4.3 are evaluated at one-second resolution. A commit included late in a second carries that second's timestamp, so the real time between its inclusion and the report that fills it is only guaranteed to exceed $\delta - 1\,\mathrm{s}$. The commit delay must therefore exceed the real-time oracle latency by at least one second, which the reference value $\delta = 2\,\mathrm{s}$ does for latencies up to one second.

13.3 Keepers and bounties

Settlement, conditional execution, breach proofs and epoch updates are permissionless. Each account's escrow pays a fixed reward per settled order and the breach bounty b_{br}; the sustainability fund tops up rewards when gas prices rise. A protocol-operated keeper provides a liveness backstop but holds no special permissions. Accounts are limited to n_{pos} open markets, so that equity evaluation, and hence every settlement and breach proof, costs $O(n_{\mathrm{pos}})$ report verifications.

13.4 Accounts, sessions and gas

Traders interact through smart accounts (ERC-4337 [8]) or delegated externally owned accounts (EIP-7702 [9]) with *session keys* scoped to Commit and Cancel on the trader's own accounts. Both are available on Sonic: EIP-7702 since release v2.1 [51], and ERC-4337 through deployed EntryPoint contracts with commercial bundler and paymaster services [32]. Session keys cannot move USDC, make claims or change the payout address, so a compromised session key can at worst place trades. Gas is sponsored from $\mathrm{V}_{\mathrm{sus}}$, through an ERC-4337 paymaster or through Sonic's protocol-level gas subsidies, which charge a sponsor's deposit for transactions to chosen contracts or functions and, since release v2.2, can charge it only for transactions that succeed [56, 57]; traders never need to hold S. Because sponsored transactions carry no priority tip and so have the lowest transaction-pool priority under load [56], Fairgame sponsors only trader-signed transactions such as Commit, whose inclusion delay affects latency but not fairness; keepers pay for their own transactions out of their rewards and may add tips. Under FeeM, 90% of the fees for gas consumed in Fairgame's own registered contracts currently returns to the protocol, which offsets part of this cost [52] (Section 6). Payouts are made in native USDC, and CCTP V2 lets traders move them to other chains [13].

13.5 Indexing and interfaces

Every transition emits events from which an indexer reconstructs the full state history. A public *firm-health* page displays $\mathrm{V}_{\mathrm{res}}$, $\mathcal{L}^0$, $\hat{\mathcal{L}}$, C_k, ℓ_k, ψ_k per rule set and market, C_k^{st}, the queue, the menu and current prices. Before purchase, the trading interface shows the farm-proof price and the zero-skill pass bound of every rule set.

13.6 Verification strategy

A reference simulator implements the engine, the valuation layer and the controller off-chain. It serves three purposes: calibrating parameters against historical prices and synthetic trader populations (zero-skill bots, hedged coalitions, skilled traders and momentum herds); differential testing of the contracts against the reference model; and stress-testing the controller. The invariants of Section 12.2 are encoded as property-based tests, and the value-moving logic (waterfall, claims and vault accounting) is targeted for formal verification before mainnet.

13.7 Reference parameters

Table 6 gives a reference parameterization for the first deployment. The values are proposals to be calibrated with the reference simulator.

14 Regulatory considerations

This section identifies questions for counsel; it is not legal advice. Depending on the jurisdiction, paid evaluations with monetary rewards may be analyzed as games of chance, as financial instruments or derivatives (in particular contracts for difference), as investment services, or as offers of crypto-asset services. Within the European Union, the Markets in Crypto-Assets Regulation

Table 6: Reference parameters for the first deployment (proposals).

Parameter	Value	Note
Markets	BTC, ETH, S, SOL	crypto only, trading 24/7
Commit delay δ	2 s	at least 1 s above measured oracle latency (whole-second timestamps)
Freshness Δ_f and synchronization Δ_{sync}	2 s and 1 s	
Oracle agreement ε_x	0.5%	per market
Base half-spread $s_{0,j}$	$\max(2\,\text{bp}, v_j(2\,\text{s}))$	conservative σ_j per market; 2 bp covers $\sigma_j \le 99\%$, 3.5 bp covers $\sigma_j \le 174\%$
Leverage cap λ	5	
Profit split α	0.8	
Epoch length	1 day	
Coverage target $C^\star$ and upper band $C^{\max}$	1.5 and 2.5	
Target loss ratio $\ell^\star$	0.6	
Loss-ratio gain η_ℓ	0.8	$\eta_\ell \ell^\star = 0.48$: monotone regime
Rate limit χ	$\ln 1.05$	at most 5% per day
Stress moves γ_j	15% (BTC, ETH); 25% (S, SOL)	over the claim horizon
Stress buffer β	0.5	

(MiCA) [20], which has applied in full since December 30, 2024 (its titles on asset-referenced and e-money tokens since June 30, 2024; Article 149), is relevant to the last of these. MiCA defines a crypto-asset as a digital representation of a value or of a right that is able to be transferred and stored electronically using distributed ledger technology or similar technology (Article 3(1)(5)), and its recitals state that crypto-asset services provided "in a fully decentralised manner without any intermediary" should not fall within its scope (Recital 22). A protocol in which a multisig proposes parameter changes at launch and a guardian can pause activity (Section 11) should not assume that this exclusion applies.

The design choices of this paper are compatible with a conservative posture but do not settle these questions. svUSD is non-transferable and non-redeemable (Definition 2.7); payouts are performance rewards computed by a published formula; evaluations are priced on a skill-neutral basis; and the protocol never takes custody of trader funds beyond the evaluation fee. Decentralized governance does not by itself shield a protocol or its contributors from liability. In a case brought by the Commodity Futures Trading Commission (CFTC), a United States district court held that the Ooki DAO, governed by token-holder votes, was an unincorporated association and therefore a person that could be sued and held liable under the Commodity Exchange Act, and entered a default judgment against it [61]. In a related settled order, the CFTC found two founders of the DAO's predecessor company personally liable for the DAO's violations as members of that association [16].

Before mainnet, the project should obtain jurisdiction-specific opinions, define the legal wrapper of the treasury and of its contributors, decide on access restrictions, and screen payout addresses against sanctions lists.

15 Roadmap

Phase 0. Research. Build the reference simulator; calibrate rule sets, spreads and controller gains against historical prices and synthetic trader populations; review this paper.

Phase 1. Testnet. Deploy the core contracts of Table 5, excluding later-phase modules, together with keepers, the indexer, the trading interface and the firm-health page; begin audits.

Phase 2. Guarded mainnet. Launch tournaments (Section 9); then open a small menu of tight rule sets with hard caps on funded capacity; register for FeeM and apply for ecosystem grants.

Phase 3. Self-balancing. Switch on the controller; introduce optional proof of personhood and verified discounts, the passport, scaling and direct funding, and the referral program.

Phase 4. Autonomy. Transition to identity-bound reputation governance; launch the copy-capital vault and net-book hedging.

16 Conclusion

A permissionless prop firm cannot rely on the levers that conventional firms use to stay solvent. Fairgame replaces them with a theory and a control loop. Under the stated assumptions on oracle prices and keepers, fair execution makes every trader without an edge face a supermartingale, up to the gap for conditional orders listed below. Optional stopping turns that fact into closed-form bounds on what any strategy without an edge, or any coalition of such strategies, can extract. Those bounds set prices that need no identity checks and liabilities that can be published at every block. A feedback controller with a known stability region steers the remaining economics from realized cash flows. The result is a firm whose solvency argument is public and whose rules are the same for everyone.

Open problems.

(1) The engine-design questions of Section 12.3: inadmissible first reports, eroded stress coverage, erasure of breaches, and a complete treatment of conditional orders.

(2) Incentive-compatible keeper rewards that guarantee Assumption 4.5 under congestion.

(3) Drift-robust valuation: pricing that degrades gracefully when Assumption 2.4 fails in trending regimes.

(4) Optimal control under adverse selection, including the joint use of price, menu and capacity.

(5) Commit privacy, for example through encrypted mempools, to reduce free-riding on the copy-capital vault without weakening verifiability.

(6) Cross-chain execution for the copy-capital vault with bounded latency.

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A Notation

Symbol	Meaning
Markets, oracles and information	
$\mathcal{M}$, j	listed markets; market index
$\omega = (j, P, c, u, u^-)$	oracle observation: price, confidence half-width, publish time, previous publish time
Δ_o, Δ_f, Δ_{sync}, Δ_m	oracle latency bound; freshness window; synchronization bound; monitoring interval
$\kappa_{\max}$, ε_x	maximum relative confidence; oracle agreement tolerance
$\mathbb{F} = (\mathcal{F}_t)$, $\mathbb{G} = (\mathcal{G}_t)$	public and oracle filtrations, $\mathcal{G}_t = \mathcal{F}_{t-\Delta_o}$
σ_j, $\bar{\gamma}$	volatility of market j; bound on relative price moves between consecutive reports
Φ, ϕ	standard normal distribution function and density
k	index of an evaluation phase (Sections 4 and 5) or of an epoch (Sections 6 and 7)
Rule sets and accounts	
R, $\mathcal{R}$, $\mathcal{R}_k$	rule set; catalog; menu offered in epoch k
S, S_f	evaluation and funded balances
K, T_k, D_k, L	number of phases; phase target; phase maximum loss; daily loss limit
D_f, L_f	funded maximum loss and daily loss limit
ξ, $D' = D + \xi$	overshoot at failure; effective maximum loss
β_{BGK}	overshoot constant of a discretely monitored Brownian barrier, ≈ 0.5826
α	profit split
λ, n_{pos}	leverage cap; maximum number of open markets
$g_{\min}$, ρ	minimum claim; fee refund fraction
B_a, A_a, E_a, N_a, Lev_a	balance, accrued crowding fees, equity, notional and leverage of account a
$q_{a,j}$, $\bar{p}_{a,j}$	position and average entry price
f_a, f_a^{day}, B_a^0, d_a	static floor; daily floor; day-start balance; its UTC day
W_a, G, t_a	cumulative gross claims; gross claim amount; last settlement time
x_{inst}, Δ_{ch}	instant-claim threshold; challenge window for larger claims
esc_a, e, b_{br}	account escrow; escrow per account; breach bounty
Execution	
o, κ, p^{trg}, t_c, δ	order; kind; trigger price; commit time; commit delay
$u^\star(\mathsf{o})$	deterministic fill time of a market order
s_j, $s_{0,j}$, κ_c, Λ_j	half-spread; base half-spread; confidence multiplier; virtual depth
φ_j	trading fee rate
ι_j, r_j, $\bar{r}_j$, θ_j	open-interest imbalance; crowding fee rate; its maximum; its threshold
$v_j(w)$	selection value over a window of length w
Valuation	
u_n, c_n	mark-to-market report times; breach-check times
$\bar{p}_k$, $\bar{p}(R)$	zero-skill pass bounds per phase and per evaluation
V_f, $V^\star$, $F^\star$	zero-skill funded value; zero-skill evaluation value; farm-proof price
r_{ref}, c_{op}, F_a	referral share; per-account operating cost, including the escrow; fee paid for account a
$\pi_a(t)$	pass bound of an evaluation account at time t
$\mathcal{L}_a$, $\mathcal{L}^0$, $\hat{\mathcal{L}}$, $\hat{\psi}$	account liability; zero-skill liability; adjusted liability; skill adjustment
Treasury and controller	
V_{res}, V_{ins}, V_{sus}, V_{dev}, V_{burn}	payout reserve; insurance; sustainability; development; burn allocation

Symbol	Meaning
$w_{\text{res}}^{\min}$, w_{ins}, w_{sus}, w_{dev}, w_{burn}, w_{sw}	waterfall weights and sweep fraction
$\mathrm{V}_{\text{ins}}^{\star}$, k_{ins}	insurance target and its multiple of L^{st}
C_k, $C^{\star}$, $C^{\min}$, $C^{\max}$	coverage; target; menu floor; upper band
ℓ_k, $\ell^{\star}$	loss ratio; target loss ratio
ψ_k	skill ratio
ν, ε_{ν}	payout intensity; its elasticity with respect to price
m_k, $z_k = \ln m_k$	price multiplier and its logarithm
η_{ℓ}, η_C, η'_C, η_Q, χ	controller gains; per-epoch rate limit
$\bar{v}_k$, $\bar{v}_{\max}$, v_{floor}	menu bound on $V^{\star}$; its maximum; its floor
Q_k, $\mathcal{Q}$	queue length; funding queue
L^{st}, C_k^{st}, Γ, γ_j, β	stress liability and coverage; immediately claimable value; stress move and buffer
Extensions and governance	
d_v, $m_v^{\min}$, n_v	verified discount factor; its floor; per-identity cap per epoch
PSR, DSR	probabilistic and deflated Sharpe ratios
b, ζ, π_r, n_{aw}	tournament buy-in; prize share; rank weights; number of funded awards
h_i, μ_i, κ_i	copy weight; mean simulated profit per cycle; execution cost
v_i, P_i, τ_i, Δ_{pause}	voting weight; cumulative payouts; funded tenure; maximum pause
$\mathcal{S}$, Υ	global state; state transition function

B Worked examples

Zero-skill values. Table 8 evaluates (3) for representative rule sets with $\alpha = 0.8$ and $S = S_f$, ignoring overshoot, costs, referrals and refunds, in which case $F^{\star} = V^{\star}$. Two-phase rule sets use the same maximum loss in both phases and in the funded stage.

Table 8: Zero-skill values of representative rule sets ($\alpha = 0.8$, $\xi = 0$, $c_{\text{op}} = r_{\text{ref}} = \rho = 0$).

Rule set	S	Targets	Max loss	$\bar{p}$	V_f	$F^{\star} = V^{\star}$
Tight	\$10,000	8%	3%	0.2727	\$240	\$65.45
Standard tight	\$25,000	10%	3%	0.2308	\$600	\$138.46
Standard	\$25,000	10%	5%	0.3333	\$1,000	\$333.33
Wide	\$50,000	10%	6%	0.3750	\$2,400	\$900.00
Two-phase	\$100,000	8%, 5%	5%	0.1923	\$4,000	\$769.23
Two-phase wide	\$100,000	10%, 5%	10%	0.3333	\$8,000	\$2,666.67

The last row uses loss limits typical of conventional two-phase evaluations. Its farm-proof price is several times the price at which evaluations with similar parameters are commonly sold, which quantifies lever (i) of Section 1: conventional pricing presumes that customers under-perform zero skill by a wide margin. Under farm-proof pricing, affordability comes from tighter rules instead (Figure 5).

Refunds. With a full refund at the first claim ($\rho = 1$), the standard rule set requires $F^{\star} = 333.33/(1 - 1/3) = \$500$.

Liability of live accounts. A funded standard account ($S_f = \$25{,}000$, $D'_f = 5\%$, so $S_f(1 - D'_f) = \$23{,}750$) at equity \$26,000 carries zero-skill liability $0.8\,(26{,}000 - 23{,}750) = \$1{,}800$; at equity \$24,000 it carries \$200. An evaluation account under the same rule set at equity \$26,500 has pass bound $\min\left(1, (26{,}500 - 23{,}750)/(0.15 \cdot 25{,}000)\right) = 0.733$ and liability $0.733 \cdot \$1{,}000 = \$733$.

Spread floor. Annualizing over a continuous-trading year of 365.25 days (31,557,600 s), a volatility of 60% corresponds to $\sigma_j = 0.60/\sqrt{31{,}557{,}600} \approx 1.07 \times 10^{-4}$ per $\sqrt{\text{s}}$, so $v_j(2\,\text{s}) \approx 1.07 \times 10^{-4}\sqrt{4/\pi} \approx 1.2$ basis points, where the neglected terms of Proposition 5.3 are of order 10^{-8}. At 100% volatility the same window gives 2.01 basis points, so a floor of 2 basis points covers volatilities up to about 99.6%. The realized volatility of smaller assets such as S can exceed 100%, hence the per-market floor $\max(2\,\text{bp}, v_j(2\,\text{s}))$ of Table 6: 150% gives 3.01 basis points, and 3.5 basis points cover volatilities up to about 174%.

Overshoot. With $\lambda = 5$, $D = 5\%$, $\sigma = 60\%$ per year and $\Delta_m = 2\,\text{s}$, Proposition 5.14(b) gives $\xi \approx 5 \cdot 0.95 \cdot 0.5826 \cdot 1.07 \times 10^{-4} \cdot \sqrt{2} \approx 4.2 \times 10^{-4}$, that is, 0.04% of S.

Controller gains. With $\ell^\star = 0.6$ and $\eta_\ell = 0.8$, $\eta_\ell \ell^\star = 0.48$ lies in the monotone regime of Proposition 7.2, and the global condition of Proposition 7.3, $\eta_\ell \nu \le 1$, holds for every population with payout intensity $\nu \le 1.25$.

C Proofs

Proof of Proposition 5.3. Under the martingale null with log-Brownian prices, $Y_s = \ln\left(P(u_0 + s)/P(u_0)\right) = \sigma B_s - \sigma^2 s/2$ for a standard Brownian motion B. Let $M_w = \max_{s \le w} B_s$; by the reflection principle M_w has the law of $|B_w|$, so $\mathbb{E}[M_w] = \sqrt{2w/\pi}$.

Write $a = \sigma\sqrt{w}$. A seller who selects the highest report obtains $I_s = \max_{s \le w} e^{Y_s} - 1 \le e^{\sigma M_w} - 1$. Since $\mathbb{E}[e^{\sigma M_w}] = \mathbb{E}[e^{\sigma |B_w|}] = 2e^{a^2/2}\Phi(a)$, we get $\mathbb{E}[I_s] \le 2e^{a^2/2}\Phi(a) - 1 = v_j(w)$, and $v_j(w) = a\sqrt{2/\pi} + a^2/2 + O(a^3)$. The inequality is strict because $Y_s < \sigma B_s$ for $s > 0$. Integrating the known law of the running maximum of a Brownian motion with drift $-\sigma/2$ gives the exact value $\mathbb{E}[I_s] = (2 + a^2/2)\,\Phi(a/2) + a\,\phi(a/2) - 1 = a\sqrt{2/\pi} + a^2/4 + O(a^3)$, where ϕ is the standard normal density; it is attained by a seller who observes the continuous path.

A buyer who selects the lowest report obtains the relative improvement $I_b = 1 - \min_{s \le w} e^{Y_s}$. Since $1 - e^y \le -y$, we have $I_b \le \max_{s \le w}(-Y_s) \le \sigma \max_{s \le w}(-B_s) + \sigma^2 w/2$, and $\max_{s \le w}(-B_s)$ has the law of M_w, so $\mathbb{E}[I_b] \le a\sqrt{2/\pi} + a^2/2$. The function $h(a) = 2e^{a^2/2}\Phi(a) - 1 - a\sqrt{2/\pi} - a^2/2$ satisfies $h(0) = 0$ and $h'(a) = a(2e^{a^2/2}\Phi(a) - 1) \ge 0$ for $a \ge 0$, so $\mathbb{E}[I_b] \le v_j(w)$ as well.

Reports arrive on a discrete grid, so the best available report is never better than the best point of the continuous path; the bounds therefore hold for any report schedule, and the leading term is attained as the grid becomes dense. □

Proof of Proposition 7.3. Let $g(z) = z + \eta_\ell(\nu e^{-z} - \ell^\star)$. For $z \ge z_{\min}$, $g'(z) = 1 - \eta_\ell \nu e^{-z} \in [1 - \eta_\ell \nu e^{-z_{\min}}, 1) \subseteq [0, 1)$, so g is non-decreasing on $[z_{\min}, \infty)$, while $g(z) - z = \eta_\ell(\nu e^{-z} - \ell^\star)$ is strictly decreasing with unique zero $z^\star$.

Suppose $z_k < z^\star$. Then $g(z_k) - z_k > 0$, and monotonicity of g gives $g(z_k) \le g(z^\star) = z^\star$. The rate-limited step yields $z_k + \min\left(\chi, g(z_k) - z_k\right) \in (z_k, z^\star]$, which projection onto $[z_{\min}, z_{\max}]$ leaves unchanged because $z_{\min} \le z_k$ and $z^\star \le z_{\max}$. Hence $z_{k+1} \in (z_k, z^\star]$. Symmetrically, $z_k > z^\star$ implies $z_{k+1} \in [z^\star, z_k)$. The sequence is therefore monotone and bounded, and converges to some $\bar{z}$. Since the increments $z_{k+1} - z_k = \text{clip}_{[-\chi,\chi]}\left(g(z_k) - z_k\right)$ tend to zero, continuity gives $g(\bar{z}) = \bar{z}$, so $\bar{z} = z^\star$. Without the rate limit, the same argument applies with $\chi = \infty$.

If $\nu \le \ell^\star$, then for $z \ge z_{\min} = 0$ we have $g(z) - z = \eta_\ell(\nu e^{-z} - \ell^\star) \le \eta_\ell(\nu - \ell^\star) \le 0$, so the projected iterates are non-increasing and bounded below by $z_{\min}$, and they converge. A limit $\bar{z} > z_{\min}$ would satisfy $\nu e^{-\bar{z}} = \ell^\star$, which is impossible because $\nu e^{-\bar{z}} < \nu \le \ell^\star$. Hence $z_k \to z_{\min}$. □